

Discriminating Because Others Might: Anticipating Third-Party Discrimination

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Abstract

Research on discrimination typically focuses on people’s unfavorable beliefs or preferences towards marginalized groups. At the same time, there is growing evidence that beliefs about third-party behaviors, independent of personally held attitudes or tastes, can have upstream consequences on decisions. In two preregistered experiments with online samples—a natural experiment using the 2024 U.S. presidential election and a controlled online experiment—I show that beliefs about third-party discrimination have a substantial impact on direct discrimination: people discriminate not based on their own preferences and beliefs, but on their (potentially biased) perceptions of how much others will discriminate. Following the 2024 U.S. presidential election, hiring managers in states where Donald Trump won the highest vote share hired fewer female workers after the election than before, while managers in states where Kamala Harris won the highest vote share hired *more* female workers after the election. This pattern was driven by increases in perceived third-party discrimination in states where Trump vis-a-vis Harris won the most votes. In the controlled experiment, directly manipulating beliefs about third-party discrimination improved outcomes for marginalized group members. Across both studies, decision makers systematically overestimated third-party discrimination, which exacerbated their own direct discrimination. Together, these results suggest that beliefs about third-parties are an important driver of discrimination independent of directly held beliefs about stereotyped groups. They also show that large-scale social events, like elections, can impact discrimination through the previously overlooked channel of changing these beliefs.

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1 Introduction

Discrimination based on gender, race or other immutable characteristics occurs across a range of organizational contexts and contributes to persistent gaps in hiring, promotion, and wages for women and racial minorities (Bertrand & Duflo, 2017; Gaddis, Larsen, Crabtree, & Holbein, 2021; Hangartner, Kopp, & Siegenthaler, 2021; Kline, Rose, & Walters, 2022). Existing research has primarily focused on two channels: discrimination driven by preferences about specific groups (i.e., taste-based discrimination) and discrimination driven by beliefs about group members’ attributes, such as their productivity (i.e., statistical discrimination).¹ This paper considers the possibility that people discriminate not because of their own preferences or beliefs about the characteristics of some group, but based on beliefs about discrimination by *others*. For example, a business owner might not hire Black workers if he thinks customers won’t frequent his business when they see a Black employee behind the counter instead of a White one. An executive recruitment firm might not propose a woman candidate for CEO if they believe shareholders are prejudiced against female executives. A hiring manager may not hire women because he believes their supervisor will invest less in their development than similar male hires. While previous work suggests such “third-party” discrimination may play an important role in labor markets (Abraham, 2020; Becker, 1993; Fernandez-Mateo & Fernandez, 2016; Holzer & Ihlanfeldt, 1998; Hurst, Rubinstein, & Shimizu, 2024; Leonard, Levine, & Giuliano, 2010; Neumark, Bank, & Van Nort, 1996), there is little direct causal evidence of its existence or significance.

Beliefs about third-party actions have been shown to affect a range of important behaviors, such as financial investing, energy consumption, alcohol consumption, and domestic violence (Jachimowicz, Hauser, O’Brien, Sherman, & Galinsky, 2018; Tankard & Paluck, 2016). This occurs both because beliefs that third parties engage in a behavior might make that behavior “normative”, encouraging norm-consistent behavior, and because third-party behaviors can directly affect decision-makers, incentivizing them to anticipate and account for those behaviors (Chang, Milkman, Chugh, & Akinola, 2019; Cialdini & Goldstein, 2004; Tankard & Paluck, 2016). For example, if an organization’s

¹See Guryan and Charles (2013), and see Bohren, Haggag, Imas, and Pope (2023) for a recent review.

clients, shareholders, or employees discriminate against members of a certain group (e.g., women), hiring or promoting members of that group could be costly for the organization, as discrimination against such employees could lead to reduced sales, profits, share price, or other costs (Kelley, Lane, Pecenco, & Rubin, 2024; Leonard et al., 2010). Managers could thus face incentives not to hire or promote people from these groups. As such, these third-party beliefs could have important implications for labor market discrimination. However, it remains unclear whether organizational decision makers anticipate such third-party discrimination, what factors affect these beliefs, and how these beliefs translate into labor market outcomes for marginalized groups.

The present work investigates these questions by both directly manipulating third-party beliefs in a controlled experiment and by leveraging a natural shock to these beliefs. This paper also focuses on the *accuracy* of these beliefs. It would be one thing if people acted on accurate beliefs about third-party discrimination—preferring candidates from a certain group because it would be costly to do otherwise—but beliefs about third-party discrimination can also be inaccurate. Indeed, research on social norms and pluralistic ignorance suggests that beliefs about others’ attitudes, beliefs and behaviors can be systematically wrong (Bursztyn, Cappelen, Tungodden, Voena, & Yanagizawa-Drott, 2023; Prentice & Miller, 1996). People tend to have incomplete information about what others do or think, which is likely to be more pronounced for sensitive topics like discrimination (Tankard & Paluck, 2017). Research on the “bias blindspot” effect also shows that people typically overestimate beliefs and actions considered “biased” (Pronin, Gilovich, & Ross, 2004; Pronin, Lin, & Ross, 2002; Scopelliti et al., 2015). Bursztyn et al. (2023), for example, found that people systematically overestimated opposition to female labor force participation. If decision makers are upwardly biased in their beliefs about third-party discrimination, then acting on these beliefs might exacerbate, rather than simply mirror or reproduce existing discrimination. In fact, if no such discrimination by third parties exists, but decision makers believe it does, this could lead to a special form of “self-fulfilling prophecy,” whereby discrimination only occurs because of these incorrect beliefs.

In this paper, I test for “third-party discrimination”—whether decision makers discriminate because of their beliefs regarding how *others* discriminate—and highlight the role of inaccurate beliefs in generating a “self-fulfilling prophecy.” I also show that aggregators of social information, e.g.,

results of a national election, affect these third-party beliefs, and demonstrate their downstream consequences on behavior. I investigate these questions using both a natural experiment that leverages the 2024 U.S. presidential election, and a controlled experiment in a stylized online labor market. I first measure and utilize the shift in beliefs about third-party gender discrimination induced by Donald Trump’s election victory over Kamala Harris in November 2024. Previous work shows that public signals, like national election results, can affect related third-party beliefs.² I propose that learning about a Donald Trump election victory, vis-a-vis a Kamala Harris victory, affected decision makers’ beliefs about third-party gender discrimination (I explain the rationale behind this proposition in more detail in Section 2.1.5). This, in turn, could affect their choices to discriminate directly, such as whether to hire a female worker for a client-facing role. To identify the effect of the election on these beliefs and discrimination, I measured beliefs and hiring choices immediately before and after the election with participants from states where the election outcome was ex ante uncertain. I then exploited variation in both which candidate won the election across different states and in the “surprisingness” of the result to individual participants within states, testing how this predicted changes in third-party beliefs and hiring choices over the election period. I find that the election meaningfully shifted participants’ beliefs about gender discrimination by others in their state: in states where Trump won, hiring managers’ beliefs about client discrimination against female workers significantly increased. Hiring managers in these “Trump States” responded by hiring significantly fewer female workers after the election than before, particularly compared to managers in “Harris States.” This result was mostly driven by those who did not anticipate the election outcome in their state. Hiring managers in Trump states who expected Harris to win and were thus “surprised” by a Trump victory, shifted their beliefs regarding client discrimination more than those who accurately predicted a Trump victory, and hired fewer women as a result. This pattern was reversed in states where Harris won the highest vote share.

I show that these changes in beliefs about clients’ discrimination caused changes in hiring choices and use several analyses to rule out the most plausible alternative interpretations, such as a change

²Large-scale social and political events—like Supreme Court rulings, national elections, and large-scale protests—serve as important aggregators of social information, providing decision-makers with information about the beliefs, preferences, and behaviors of others in their society. This has been shown to affect beliefs about third-party behavior, and related behavior towards marginalized groups. See, for example: Bursztyn, Egorov, and Fiorin (2020); Grosjean, Masera, and Yousaf (2023); Larrebourg and Gonzalez (2021); Müller and Schwarz (2023); Tankard and Paluck (2016, 2017).

in beliefs about male-female productivity and selection effects. However, to provide more support for the proposed mechanism, I also conducted a controlled online experiment aimed at isolating the effect of third-party beliefs directly. Here, I used the same hiring paradigm, but directly manipulated beliefs about client discrimination, holding other factors fixed. The results from this controlled experiment confirmed those of the natural experiment—hiring managers responded to signals that clients discriminate less against women by hiring more female workers. Taken together, these studies demonstrate the causal effect of beliefs about third-party discrimination on hiring decisions.

However, the critical question remains: Are hiring managers’ beliefs about client discrimination accurate? In both experiments, I find that hiring managers overestimate clients’ discrimination against female workers and consequently hire fewer female workers than they would with accurate beliefs. In fact, clients in both experiments do not discriminate against female workers, while hiring managers believe that they do, and these incorrect beliefs cause them to discriminate. This then generates discrimination in this market as a special form of “self-fulfilling prophecy”.

This paper makes several contributions. First, it builds on the literature on statistical discrimination (Arrow, 1973; Phelps, 1972). This previous work largely focused on decision makers’ beliefs about group members themselves, such as their skills or productive capacities (Barron, Dittmann, Gehrig, & Schweighofer-Kodritsch, 2024; Bohren et al., 2023; Campos-Mercade & Mengel, 2023). The present paper shows that even without negative beliefs about group members, beliefs about third-parties can cause decision-makers to discriminate, building on recent work by Bohren, Imas, and Rosenberg (2019) and Bohren, Hull, and Imas (2025). Unlike those papers, the present work focuses on the effect of anticipating future discrimination rather than on beliefs about previous discrimination. The paper also builds on previous research on customer discrimination (Holzer & Ihlanfeldt, 1998; Hurst et al., 2024; Neumark et al., 1996), showing that firms tend to favor workers with the same demographic characteristics as their clients. The present work directly manipulates beliefs about clients, rather than inferring them from their demographics, and identifies the effect of these beliefs on hiring. In this way, the present paper connects the literature on customer discrimination with that on statistical discrimination by centering third-party beliefs as a mechanism for customer discrimination.

This work also contributes to the literature on belief biases and discrimination. There has been a surge of recent work showing that motivational, perceptual, and inferential biases—such as the representativeness heuristic (Bordalo, Coffman, Gennaioli, & Shleifer, 2016; Coffman, Exley, & Niederle, 2021; Esponda, Oprea, & Yuksel, 2023), non-Bayesian belief updating (Campos-Mercade & Mengel, 2023; Coffman, Collis, & Kulkarni, 2024), and motivated reasoning (Eyting, 2022; Rackstraw, 2022; Stötzer & Zimmermann, 2024)—can cause belief distortions about members of certain groups and discrimination. This paper proposes and demonstrates that a form of “pluralistic ignorance” (Bursztyn et al., 2023; Prentice & Miller, 1996; Pronin et al., 2002), or systematic misperceptions of others’ beliefs and behaviors, can also cause biased beliefs and discrimination.

This paper also builds on recent research on prejudice accommodation (Lide & Berg, 2023; Vial, Brescoll, & Dovidio, 2019; Vial, Dovidio, & Brescoll, 2019). This prior work shows that when decision-makers receive information suggesting a manager is prejudiced (e.g., they become aware of a manager’s personal beliefs or they infer such beliefs from the degree of diversity in his firm) they “accommodate” the manager’s prejudice by hiring fewer people from the targeted group. The present paper builds on this work by testing how beliefs about discrimination by “general others” (such as a randomly drawn client) can affect candidate selection, and by showing how naturally occurring public signals (like an election) might affect these beliefs. Moreover, by testing the accuracy of these beliefs compared to ground truth, I also show that these beliefs can be systematically wrong and can exacerbate existing discrimination in settings with real monetary stakes.

Finally, this work adds to a growing literature on the effect of landmark social events—such as national elections, Supreme Court decisions, and large-scale protests—on discrimination and other inter-group attitudes and behaviors (Edwards & Rushin, 2018; Feinberg, Branton, & Martinez-Ebers, 2022; Grosjean et al., 2023; Larreboure & Gonzalez, 2021; Müller & Schwarz, 2023). The present work provides evidence of a potential causal channel for the above effects, namely that these events affected decision makers’ third-party beliefs, and consequently their own behavior, and extends the above findings to discrimination in the labor market.

2 A Natural Experiment

2.1 Design

To test for the existence and implications of third-party discrimination, I conducted a preregistered incentivized hiring experiment in a stylized online labor market ([AsPredicted #197318](#)). In this study, I used the 2024 U.S. presidential election as a natural source of variation in beliefs about third-party discrimination, and examine how those beliefs impacted gender discrimination in hiring. Participants played one of three roles in this labor market: workers (who completed a set of incentivized tasks), clients (who selected workers based on their expected task output), and hiring managers (who hired workers they believed clients preferred). Hiring managers were specifically paired with clients from their own state of residence for all choices and belief measures. This setup allowed hiring manager and client decisions to be fully incentivized—about real online workers—and free of deception. For clients and hiring managers, I measured worker selection in a hiring task and a set of beliefs twice, both (i) before the 2024 U.S. presidential election (the “baseline survey”; conducted 1-2 days before the election) and (ii) after the election (the “endline survey”, conducted 10-13 days after the election). This allowed me to measure how participants’ choices and beliefs changed over the election period. I focus my analysis on changes in hiring managers’ choices and beliefs about client discrimination. Below I give more detail on each stage of the experiment.

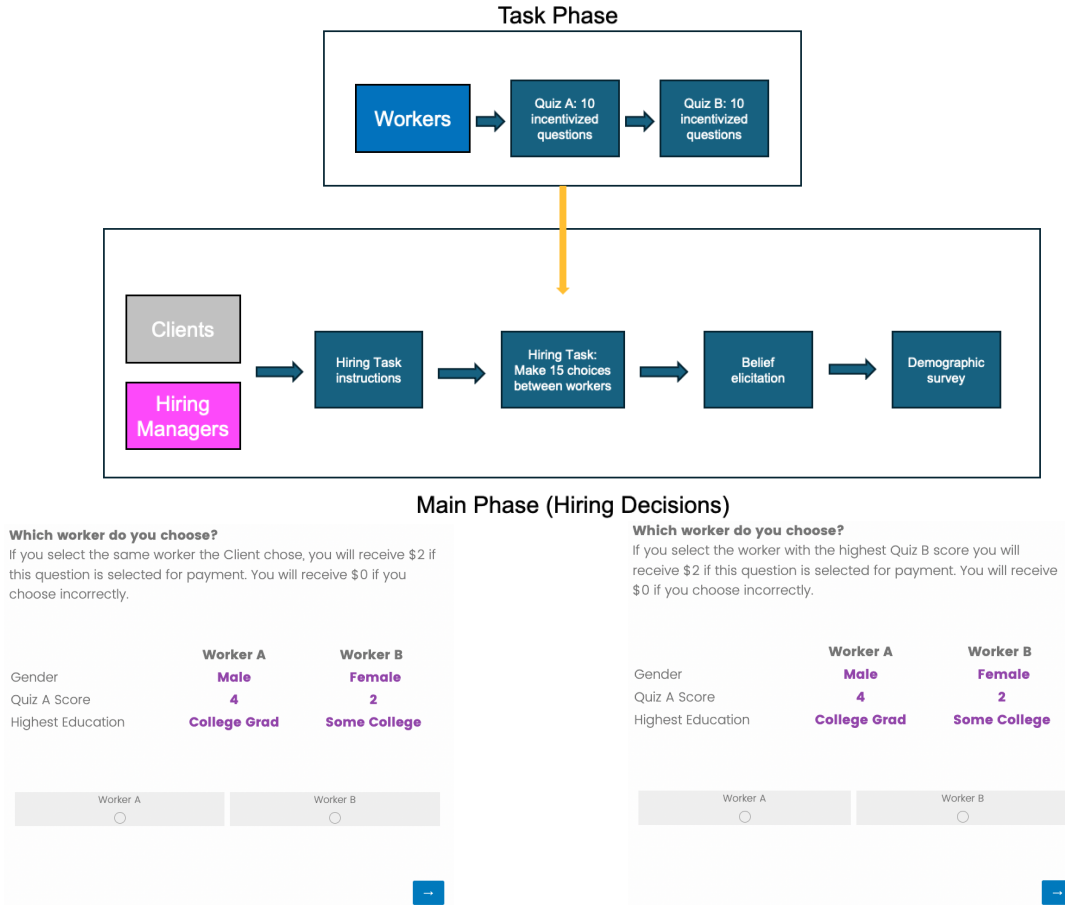
2.1.1 Hiring Task

Figure 1, panel (a) summarizes the survey flow for each participant type. In the first phase of this experiment, the “task phase”, 50 male and 50 female online workers completed two incentivized “male-stereotyped” tasks: two quizzes on math, finance and history (called “Quiz A” and “Quiz B”). People typically expect female workers to perform worse than male workers on these and similar tasks, and also expect that others believe this (Bertrand & Duflo, 2017; Bohren, Hull, & Imas, 2022; Coffman, 2014; Feld, Ip, Leibbrandt, & Vecchi, 2022; Guiso, Monte, Sapienza, & Zingales, 2008).³ In these data, however, male and female workers performed approximately equally well on both tasks (see Figure 6 for the distribution of scores in Quiz B by gender). This constituted the

³In a previous study using this identical task, Prolific participants believed male workers outperformed female workers (Bohren et al., 2022).

Figure 1: Survey Flow and Hiring Task Decision Screens

(a) Survey Flow for Each Phase



(b) Hiring Manager Decision Screen

(c) Client Decision Screen

Notes: Panel (a) shows the experimental flow for the task phase and the main phase, which was the same for both the controlled and natural experiment. Note, that clients and hiring managers completed the same flow for the main phase twice in the natural experiment: both before and after the 2024 U.S. presidential election. Panels (b) and (c) give examples of the decision screens that hiring managers and clients saw during the hiring task, respectively. They each saw 15 decision screens like this, with workers randomly drawn from the same pool.

“worker pool” for this experiment.

In a separate phase, the “main phase,” I presented both hiring managers and clients (sample descriptions in Section 2.1.4) with 15 randomly selected pairs of workers drawn from the worker pool. For each pair, hiring managers and clients observed a “productivity signal” for each worker, including their highest level of education and their score on the first of the two tasks (Quiz A). They also observed each worker’s gender (see Figure 1, panel (b) and (c) for examples of these decision screens). Hiring managers and clients saw the same decision interface and the same worker pairs, but had different objectives. For each pair, clients were incentivized to select the most productive worker in the pair based on their score on the second task (*“select the worker with the highest Quiz B score”*). Meanwhile, hiring managers were incentivized to select the worker that a randomly chosen client selected (*“select the worker that the client chooses”*). Both clients and hiring managers received a bonus if they selected the correct worker in one randomly selected worker pair. To determine their bonus, hiring managers were yoked to a randomly selected client from their state of residence. Thus, much like in many labor markets, hiring managers had an interest in hiring workers that they believed others, in this case clients, would endorse. Hiring managers and clients completed the same hiring task at both baseline (before the election) and endline (after the election).

2.1.2 Belief Elicitation

For hiring managers only, I measured beliefs about clients’ average discrimination against female workers. To incentivize the accurate reporting of beliefs, I followed the procedure in Danz, Vesterlund, and Wilson (2022) and asked managers to provide their best estimates of the average proportion of female workers that clients would select, providing them with a bonus if they guessed within 3.3 percentage points of the true value.⁴ This constituted my primary pre-registered measure of third-party discrimination beliefs. I also measured both client and hiring manager beliefs about male and female worker productivity using the same procedure and incentives.⁵ To measure hiring

⁴If this question was randomly selected for payout, they received an additional \$1.00 bonus. To check this measure, I asked them a 5-point qualitative Likert scale about how much on average clients favored male or female workers. These measures were highly correlated: $r = 0.38, p < 0.001$.

⁵I used different thresholds to determine the bonus. Those within 5 percentage points of the correct average received a \$1.00 bonus if this question was randomly selected for payout.

managers’ more general beliefs about others’ discrimination, not just those of clients in this setting, I also asked them to assess how many people out of 100 they believed supported female labor force participation (taken from the Gallup World Poll) and how supportive they believed others were (on a 7-point Likert scale) of jobs being reserved for men when they are scarce (reverse coded; taken from the World Values Survey). I collected these belief measures both before and after the election.⁶ See Appendix Supplement B.2 for all belief questions, and OSF for all materials.

Additionally, to determine the ex post “surprisingness” of the election result for hiring managers, I measured their baseline beliefs about the election outcome in their state and at the national level. For each hiring manager, in the baseline survey I recorded which candidate they believed would win, the degree of certainty in their choice (on a 7-point Likert scale), and the margin by which they believed the candidate would win (in percentage difference in vote/electoral seat share between the first and second-placed candidate). I incentivized all estimates of the election winner and the winning margin.⁷ See Appendix Table D6 for a summary of these baseline beliefs.

2.1.3 Dependent Variables

My primary pre-registered dependent variables include *changes* in hiring managers’ beliefs about client discrimination and *changes* in the decision weight hiring managers placed on worker gender in their hiring choices. For beliefs, I calculated the difference in managers’ incentivized beliefs about the proportion of female workers clients selected between baseline and endline surveys. I estimated the decision weight on worker gender separately at baseline and endline by regressing each manager’s choices in the hiring task on worker gender and qualifications (i.e. workers’ highest level of education and Quiz A scores). This captures hiring managers’ gender discrimination holding worker qualifications fixed. I then take the difference between the endline and baseline weight estimates. See the Analysis Supplement B.4 in the Appendix for more details on how these weights

⁶The one exception is that I only collected hiring manager beliefs about male and female productivity in the endline survey. I did this to limit the number of beliefs elicited during the baseline survey to avoid fatigue during responding (as hiring managers also provided a battery of beliefs about the election at baseline), which might have affected the quality of other belief measures that were first order to identification.

⁷Participants were paid a bonus if they guessed the correct winner if that question was randomly selected for payout. For the winning margin, I used the same method as for the beliefs about worker productivity and client discrimination, providing a bonus if they estimated within 3 percentage points or 10 electoral seats of the actual winning margin for the state and national levels respectively, following the incentivization procedure in Danz et al. (2022).

were derived for each worker. For ease of interpretation, I also report the proportion of female workers hired, which is highly correlated with the estimated gender weight ($r = 0.89$).

2.1.4 Sample

As pre-registered, I recruited 600 participants on Prolific Academic (500 hiring managers and 100 clients), aiming for a final sample of at least 480 participants (400 hiring managers and 80 clients) completing both baseline and endline surveys after exclusions. I restricted recruitment to participants who resided in 11 U.S. states where the election outcome was uncertain prior to the 2024 presidential election, according to poll data. This included all 7 “swing states” (where the predicted margin of victory was less than 2 percentage points), and 4 other states where the predicted margin of victory was still relatively low (see the Appendix Sample Selection supplement for details about state selection and polling data). As pre-registered, I screened out participants who failed attention checks, were living in a state outside of the specified set at the time of the surveys, or failed to complete the endline survey. After exclusions, the final sample included 497 participants (412 hiring managers and 85 clients) for the primary pre-registered analyses (see Table B2 in the Appendix for details about this sample). Given that my preregistered analysis relies on difference-in-differences for only those that completed both baseline and endline surveys, attrition should not affect the internal validity of my findings. However, I find no evidence that the final sample differed from the sample recruited at baseline (see Tables B3 and B4 in the Appendix).

2.1.5 The Election as a Belief Shock

Elections have been shown to provide public signals of what others in society think or prefer, causing people to update their beliefs about third party behavior. For instance, previous work shows that following Donald Trump’s victory in the 2016 U.S. presidential election, perceptions of his local electoral success directly impacted public support for anti-immigration policies (Bursztyn, Egorov, & Fiorin, 2020). I propose that the 2024 U.S. presidential election could have caused similar shifts in participants’ beliefs about third-party gender discrimination, driven by perceived support for one political party (and electoral candidate) compared to support for the other. Prior to the election, the Republican Party candidate, Donald Trump, had been described as “sexist,” “misogynistic,” and representative of a culture of “hypermasculinity” in the popular press, and was

the target of several allegations of sexual assault (Bidgood, 2024; Bracic, Israel-Trummel, & Shortle, 2019; Filipovic, 2017; Foreign Policy, 2024; Hesse, 2024; Neumeister, Peltz, & Sisak, 2023; Powell, Butterfield, & Jiang, 2018; Prasad, 2024; The Guardian, 2024a, 2024b). In fact, the global Women’s March in 2017 was touted as a direct protest to Trump’s election as U.S. president in 2016 (Bracic et al., 2019; Larrebourg & Gonzalez, 2021). Moreover, support for Trump in the 2016 election predicted regressive views of women in society (Setzler & Yanus, 2018) and discrimination against other marginalized groups.⁸ By contrast, Kamala Harris, the opposition and electoral candidate for the Democratic Party would have been the first female president of the United States and was considered as largely progressive regarding women’s role in society (Armour, Appleby, & Rovner, 2024; NPR, 2024).

The political parties involved in the election reflected a similar divide. As of the election in November 2024, 41% of the Democratic Party’s representatives in the U.S. Congress were women, compared with just 16% of Republican representatives, reflecting a historical pattern of differential female representation between the two parties in Congress, the U.S. Senate, and senior political positions (Pew Research Center, 2023).⁹ Finally, public discourse and the popular press consistently reinforced the role of gender in the 2024 election, including the gender gap in voting patterns that saw female voters favor the Democratic Party and male voters favor the Republican Party (Foreign Policy, 2024; Page, Hoff, & Koch, 2024; The Economist, 2024; Zhou, 2024).

Taken together, Donald Trump winning the election could have provided hiring managers with information about others’ preferences and beliefs regarding gender. As Prof. Laura Kray described in November 2024, “The results [President Trump’s victory in the 2024 election] appear to confirm that most American voters still implicitly endorse traditional gender hierarchies...” (Lempinen, Pohl, & Thulin, 2024). Given the above, I argue that learning about a Donald Trump victory could have caused people to update their beliefs about third-party gender discrimination, and consequently affected their willingness to hire a female worker. I verify these shifts in beliefs in Section 2.2.1.

⁸Related work documents increases in racial discrimination caused by public displays of support for Donald Trump (Grosjean et al., 2023), and increases in hate crimes following both his public statements and election victory in 2016 (Edwards & Rushin, 2018; Feinberg et al., 2022; Müller & Schwarz, 2023).

⁹As of 2024, the only female presidential candidates from a major political party in the U.S. were from the Democratic Party, as were the only female Vice President and Secretary of State (Pew Research Center, 2023).

2.1.6 Identification Strategy Overview

My primary pre-registered analysis tests how beliefs about third-party discrimination and the choice to hire female workers changed after the 2024 U.S. election as a function of (i) which candidate won in a hiring managers' state, and (ii) how unexpected this election result was for each hiring manager (see Supplement B.4 for details on the analysis strategy and the preregistered regression specifications). Several features of the design support identification. First, the analysis uses a difference-in-difference approach, where the primary dependent variables are changes from before to after the election: the change in manager beliefs about client discrimination, and the change in managers' likelihood of hiring a female worker. This removes the effect of time-invariant differences between hiring managers in analysis. By keeping the period between baseline and endline measurement short (12-17 days), I limited the scope for violations of the parallel trends assumption: it is unlikely that beliefs about third-party discrimination or the likelihood of hiring female workers changed over this short period for reasons unrelated to the election, and even less likely that any such changes were different for treated and untreated units (i.e., between states where one candidate vs another won the election). For robustness, I run specifications including different sets of control variables, accounting for any deviations in the parallel trends assumption that are predicted by participants' baseline characteristics, and run tests on other outcomes as a placebo test.

Second, I test for the effect of the election (as a form of belief shock about third-party discrimination) by both leveraging state-level variation in which candidate won the election, and by leveraging individual-level variation in how surprising this result was relative to hiring managers' baseline beliefs. To operationalize "surprisingness," I use deviations between the observed election outcome and participants' ex-ante predictions about the outcomes. I propose that deviations should cause belief-updating about third-party discrimination (Bursztyn, Egorov, & Fiorin, 2020). I specifically focused on states where poll data suggested the election would be closely contested in order to increase variation in the surprisingness of the result (Jennings & Wlezien, 2018; Kotak & Moore, 2022): in states where the election was closely contested, participants should have been both less certain about which candidate would win, and have been more likely to wrongly predict the winning candidate (and thus be surprised by *which* candidate won the election). I leverage both

categorical deviations from managers’ predictions (i.e., failures to predict *which* candidate would win the election), and continuous deviations in terms of the winning candidates’ vote share. I specifically prioritize the categorical surprise measure because decision-makers tend to place more weight on outcomes close to, and that move across, some reference point or boundary (Esponda et al., 2023; Naborn & Bogard, 2025; O’Donoghue & Sprenger, 2018). For example, learning that Kamala Harris received a 48% vote share and *lost* the state, instead of winning with an anticipated 51% share, might cause a manager to update their beliefs about others more than if Harris received 65% of the vote instead of an anticipated 68% and still won the state. In both cases, the deviation is 3 percentage points, but in the former case a manager might pay more attention to this deviation and update their beliefs more. For each analysis presented below, I describe which independent variable I used in each case.

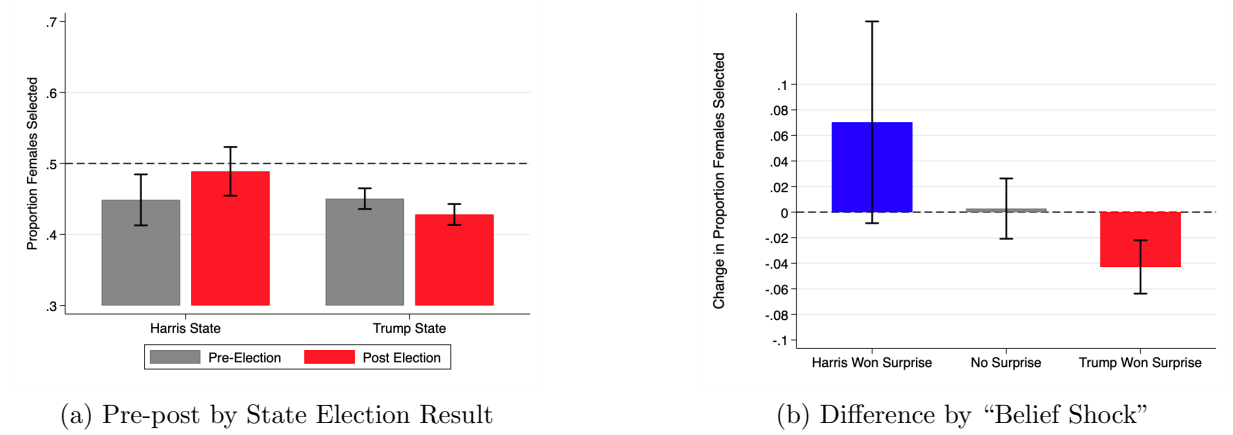
2.2 Results

2.2.1 The Election Shifted Beliefs about Third-Party Discrimination

First, I test whether the election outcome affected beliefs about third-party discrimination. Figure 2 Panel (a) shows hiring managers’ beliefs about the proportion of female workers that clients (in their same state) selected in the hiring task, split by the type of state (i.e., states where Trump or Harris eventually won) and split by pre- and post-election. Figure 2 shows that before the election, hiring managers in both states where Trump eventually won the election (Trump states) and states where Harris eventually won (Harris states) held similar beliefs about third-party discrimination: both groups expected clients in their states to choose fewer female than male workers (selecting female workers approximately 45% of the time, significantly less than parity in a one-sample t-test: $\Delta = -0.050 : t = -7.23, p < 0.001$). However, after the election, these beliefs diverged. Managers in Trump states became more pessimistic about clients’ gender discrimination—the expected rate of female worker selection dropped from 45.0% before the election to 42.8% after the election ($t = -2.53, p = 0.012$). By contrast, managers in Harris states became more optimistic—the expected rate of female worker selection increased from 44.9% to 48.9% for managers in these states (though this change was not significant at $\alpha = 0.05$: $t = 1.98, p = 0.053$). There was thus a difference in these belief *changes* between managers in Trump compared to Harris states of 6.2 percentage points

($t = 2.83, p = 0.005$). Pre-registered regressions confirm this result, and show that it is robust to the inclusion of several sets of control variables (see Appendix Table D7). Managers' beliefs about client gender discrimination thus responded differently depending on the election outcome in their state.

Figure 2: Hiring Manager Beliefs about Clients' Choices: Proportion of Female Workers Clients Selected



Notes: Panel (a) shows the mean hiring manager belief about the proportion of female workers clients selected, separately for (i) U.S. states in which Trump won the highest share of votes in the 2024 National Election and states in which Harris won the highest share, and (ii) pre-election (baseline) and post-election (endline) measures. The dotted line represents 50% male and female workers hired. Panel (b) shows the mean values for the change in these beliefs from baseline to endline (the change in the proportion of female workers that hiring managers believed clients selected). Positive values indicate that hiring managers' estimates of the number of women that clients selected increased after the election (relative to before), and negative values indicate these estimates decreased. Zero reflects no change in managers' beliefs about the number of women that clients selected. This is displayed separately for hiring managers that (i) failed to predict a Harris victory in their state (i.e., Harris won, but they predicted Trump would win), (ii) correctly predicted the winning candidate, (iii) failed to predict a Trump victory in their state. Error bars represent 95% confidence intervals of the means.

Next, I investigate how these changes in beliefs varied as a function of managers' baseline beliefs about who would win the election (and thus, how surprising the election result was for each manager). Figure 2, Panel (b) shows the change in managers' beliefs about clients' female worker selection over the election period, split by whether they were "surprised" by a Trump victory (i.e., they predicted that Harris would win in their state, but Trump eventually won), surprised by a Harris victory (i.e., they predicted Trump would win in their state, but Harris eventually won), or they correctly predicted the election outcome. Figure 2 shows that the changes in beliefs about clients' gender discrimination were largest for hiring managers whose baseline beliefs about the election winner in their state were incorrect. Hiring managers surprised by a Trump victory in their state reduced their beliefs about the proportion of female workers that clients selected by 4.3 percentage points ($t = 4.03, p < 0.001$). Hiring managers surprised by a Harris victory expected

clients to hire 7.0 percentage points *more* female workers after the election than before, though this was not significant at $\alpha = 0.05$ ($t = 1.75, p = 0.093$). There is no evidence that hiring managers who correctly predicted the election outcome changed their beliefs about clients' female worker selection ($\Delta = 0.3\%, t = 0.23, p = 0.82$). Pre-registered regressions confirm that this categorical surprise about the election winner (i.e., failing to anticipate *who* would win the election in their state) significantly affected hiring managers' beliefs about clients' gender discrimination, including after controlling for which candidate actually won in each state (see Appendix Table D7 for these regression results, and column 5 for those including state fixed effects).

Additionally, I pre-registered testing a second measure of how surprising the election result was: the difference between managers' baseline beliefs about the winning candidates' vote share in the election and the candidate's actual vote share (i.e., a continuous measure of how surprising the election result was given their baseline beliefs). This measure also significantly predicted hiring managers' changes in beliefs about clients' gender discrimination (Appendix Table D7). Taken together, managers who failed to correctly predict the election outcome, both the winner and the winning candidates' vote share, and thus for whom the election was more informative, changed their beliefs about client discrimination more than those who accurately predicted the election result.

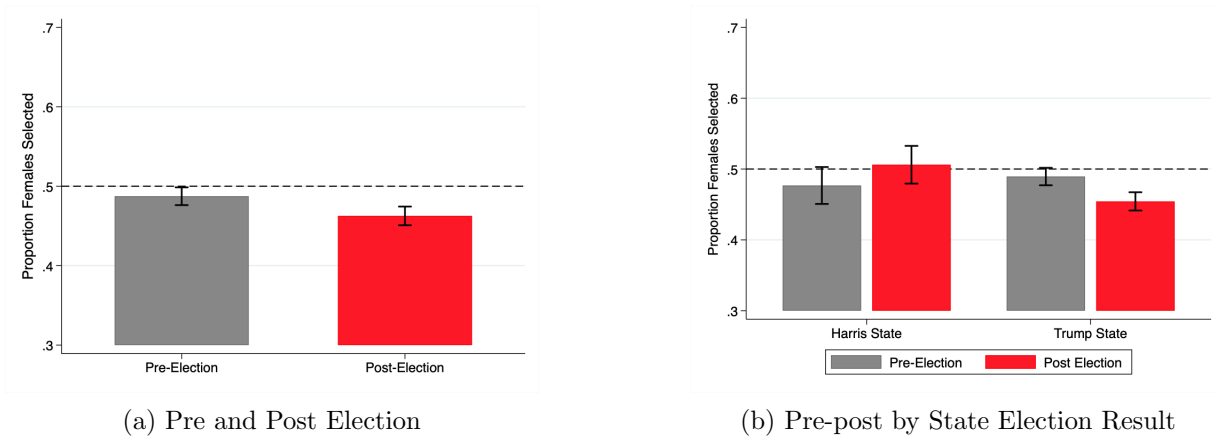
2.2.2 The Effect on Hiring Discrimination

The Effect of the Election on Hiring

In the above subsection I show that hiring managers expected clients to discriminate against female workers more after the election than before in Trump States, and less in Harris states. How do managers' hiring decisions correspond to these belief changes? First, Figure 3, Panel (a) shows that on average hiring managers across all surveyed states hired fewer female workers after the election than before ($\Delta = 2.48\%, t = 3.53, p < 0.001$). Figure 3, Panel (b) then shows the proportion of female workers that hiring managers selected, split by both pre- and post-election and by which candidate won the election in their respective state. Like with beliefs, the overall reduction in female worker hiring was driven by hiring managers in states where Trump won the highest vote share. Hiring managers in these states hired 3.5% fewer female workers after the election than before ($t = 4.61, p < 0.001$), while those in states where Harris won hired 2.9% *more* female workers than

before ($t = 1.76, p = 0.082$; Figure 3). Figure 4 Panel (a) shows the same pattern of results for my primary pre-registered dependent variable—the change in the weight managers placed on worker gender in their hiring choices, holding worker qualifications (highest education level and Quiz A score) fixed (see Section 2.1.3 for a description of this variable). Managers in Trump states were less likely to hire female workers (relative to male workers) after the election than before ($p < 0.001$), holding worker qualifications constant, while this likelihood increased for managers in Harris states ($p = 0.017$); the difference in this weight change between Trump and Harris states was highly significant ($\beta = -0.071, t = -4.11, p < 0.001$; the results of the pre-registered regression are given in Table 1).

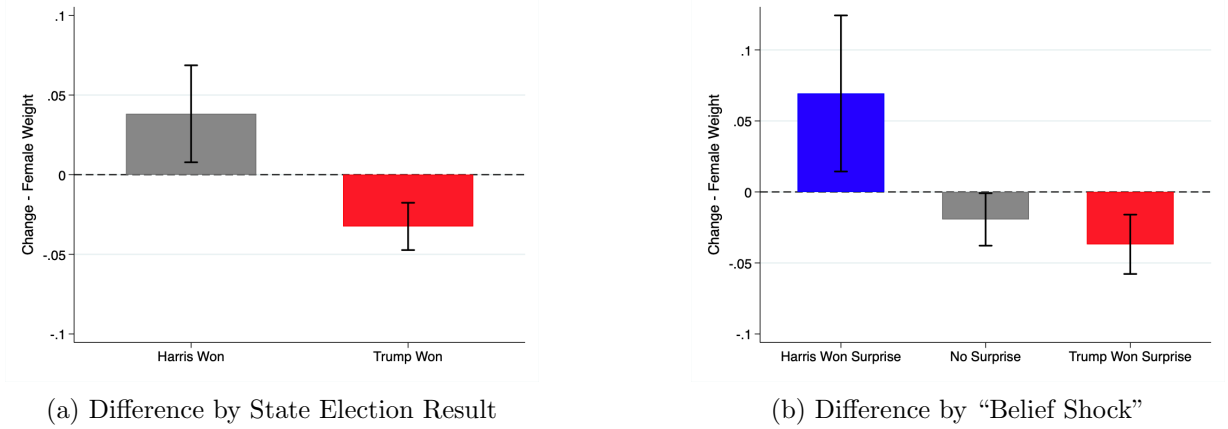
Figure 3: Hiring Managers' Hiring Choices: Proportion of Female Workers Selected



Notes: Panel a) shows the mean proportion of female workers hired by hiring managers before and after the election, across all hiring managers. Panel b) shows the mean proportion of female workers hired by hiring managers, separately for i) U.S. states in which Trump won the highest share of votes in the 2024 National Election and states in which Harris won the highest share, and ii) pre-election (baseline) and post-election (endline) measures. The dotted line represents 50% male and female workers hired. Error bars represent 95% confidence intervals of the means.

The surprisingness of the election result, given hiring managers' baseline expectations about which candidate would win the election, also affected the decision of whether to hire a female worker. Figure 4, Panel (b) shows that managers expecting Harris to win in a state where Trump eventually won were significantly less likely to hire a female worker after the election than before, while managers who failed to predict a Harris victory hired more female workers after the election. The preregistered regression shows that this failure to anticipate the election winner significantly predicted the change in a manager's likelihood of hiring a female worker ($\beta = -0.035, t = -2.95, p = 0.003$; Table 1, column 1). While this effect becomes marginally significant when I include state fixed effects in the regression ($t = -1.90, p = 0.059$), this is largely driven by decreases in the

Figure 4: Hiring Managers' Hiring Choices: Decision Weights on Gender



Notes: Panel a) shows the mean change in the decision weight that hiring manager’s placed on gender over the election (positive values indicate weighing female gender more positively after the election than before, and negative weights indicate weighing female gender more negatively after the election). These weight changes represent the change in the likelihood of hiring female workers vis-a-vis male workers, holding worker qualifications constant. Panel b) again shows the mean values for this change in decision weight, but separately for hiring managers that i) failed to predict a Harris victory in their state (i.e., Harris won, but they predicted Trump would win), ii) correctly predicted the winning candidate, iii) failed to predict a Trump victory in their state. Error bars represent 95% confidence intervals of the means.

likelihood of hiring female workers by managers who correctly anticipated Trump’s victory in their state (see Table D11). The secondary measure of “election surprise” – the difference between actual and managers’ expected margin of victory for the candidate in their state – did not significantly predict changes in hiring female workers in the main regression, though results were inconsistent across specifications (see Table 1 and Appendix Figure B1).¹⁰

The Causal Effect of Third-Party Beliefs on Hiring

I have demonstrated that the 2024 U.S. presidential election result and its surprisingness generated both changes in beliefs about third-party (client) discrimination and changes in hiring choices. To support the claim that these third-party beliefs caused the changes in hiring, I use instrumental variable (IV) regression analysis as per my pre-registration. I use the same set of independent variables – the variation in which candidate won the election in each state, and in the election “surprisingness” – as instruments for the change in beliefs about client discrimination. Instrumental variable regression requires that the instruments both predict the independent variable, i.e., changes in third-party discrimination beliefs (the “first-stage” of the IV regression), and do not affect the

¹⁰This analysis is sensitive to outliers in managers’ predictions about the size of the margin of victory in the election. When removing outliers or using an alternative measure of certainty to calculate marginal surprise (a Likert scale of how certain they were about their prediction of the election winner) the regression coefficients for the effect of marginal surprise become significant (please see Appendix B.5 for more details).

Table 1: Regression Results for Primary Outcome,
Change in Gender Weight in Hiring, by “Belief Shock”

	1	2	3	4	5	6	7	8
Trump won state	-0.071*** (0.017)	-0.068*** (0.017)	-0.067*** (0.017)	-0.068*** (0.017)		-0.063*** (0.018)	-0.064*** (0.015)	-0.06*** (0.015)
Surprise: election winner	-0.035*** (0.012)	-0.034*** (0.012)	-0.034*** (0.012)	-0.032** (0.013)	-0.03* (0.016)	-0.029** (0.013)	-0.027*** (0.01)	-0.026*** (0.01)
Surprise: winning margin	-0.031 (0.021)	-0.035 (0.022)	-0.034 (0.022)	-0.026 (0.024)	-0.03 (0.023)	-0.025 (0.024)	-0.02 (0.018)	-0.021 (0.019)
	Main	Control	Control + Productivity belief	Control + Politics	Control + State Fixed Effects	All controls	Baseline Be- lief	Baseline Be- lief + Con- trols

Notes: Each column and row correspond to a separate OLS regression specification, with coefficients on the primary independent variable reported and robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Each row represents regressions with a different set of independent variables (different operationalizations of the “belief shock”), representing the β_1 coefficient in regression equations 1, 2, and 3 respectively (shown in the Supplement B.4). The dependent variable (in columns 1-6) is the change in the weight the hiring manager places on gender (the weight placed on the worker being female) when making hiring choices from before to after the election, calculated by subtracting the baseline from the endline gender weights, determined by hiring managers’ choice data. Control variables vary by specification as indicated, but the base controls (columns 2-6 and 8) include dummy variables for black and white race, male gender, college educated (or not), and continuous age and age squared. Productivity belief indicates incentivized beliefs about the relative male and female worker productivity, measured by predicted task scores; Politics refers to political party affiliation fixed effects, and “All Controls” represent all the above control variables, and additionally income and highest education level fixed effects. Columns 7 and 8 provide results for regressions using the alternative specification recommended by (McKenzie, 2012), which regresses endline gender weight on our key independent variables, controlling for baseline gender weight (rather using the change in gender weight as the dependent variable).

dependent variable through any channel other than these changes in third-party discrimination beliefs (the “exclusion restriction”). Section 2.2.1 shows that the elections significantly affected hiring managers’ beliefs about clients’ discrimination, fulfilling the “first-stage”. In the immediate sub-section below on robustness, I also argue that it is unlikely that the election results affected hiring discrimination through channels other than third-party beliefs (such as through selection effects, beliefs about productivity, or other changes in decision-making), making a case that the exclusion restriction also holds. In the IV regressions, I find that beliefs about clients’ discrimination significantly affected hiring managers’ own discrimination (Table 2), and that the pattern of significance is largely the same as for the direct effect of the election on hiring. This suggests that the election specifically affected hiring choices through beliefs about client discrimination, providing evidence for the causal effect of third-party beliefs on discrimination.

Robustness of the Hiring Effects

The effects of the election on female worker hiring, specifically the effect of which candidate won in a state and whether the winning candidate was predicted, are robust to different specifications and to tests of alternative explanations. First, these results are highly significant ($p < 0.004$ in the main specifications; Table 1) and relatively consistent across different types of hiring managers, in-

Table 2: Instrumental Variable Regression Results:
Causal Effect of Belief Changes on Hiring

	1	2	3	4	5	6	7	8
Trump won state	1.131** (0.488)	1.149** (0.529)	1.141** (0.524)	1.203** (0.565)		1.072** (0.509)	1.01** (0.417)	1.01** (0.443)
Surprise: election winner	0.681** (0.288)	0.689** (0.299)	0.69** (0.299)	0.704* (0.359)	0.601 (0.382)	0.635* (0.345)	0.532** (0.235)	0.527** (0.243)
Surprise: winning margin	0.471 (0.365)	0.506 (0.375)	0.496 (0.37)	0.424 (0.442)	0.436 (0.377)	0.423 (0.442)	0.302 (0.289)	0.305 (0.294)
	Main	Control	Control + Productivity belief	Control + Politics	Control + State Fixed Effects	All controls	Baseline Be- lief	Baseline Be- lief + Con- trols

Notes: Each column and row correspond to a separate two-stage-least-squares (2SLS) regression specification. The primary statistic in is the coefficient on the change in hiring managers’ beliefs about client discrimination (the primary independent variable, reported in the second-stage of the 2SLS). Thus, each coefficient represents an estimate of the causal effect of changes in beliefs about client discrimination on the likelihood of hiring a female worker. Standard errors for each coefficient in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Each row represents regressions with a different set of instruments for the change in beliefs about client discrimination, corresponding to the variables $TrumpState_i$, $ElectionSurprise_i$, and $ElectionSurpriseDelta_i$ from regression equations 1, 2, and 3 respectively as instruments (shown in the Supplement B.4). The dependent variable (Y_i in columns 1-6) is the change in the weight the hiring manager places on gender (the weight placed on the worker being female) when making hiring choices from before to after the election, calculated by subtracting the baseline from the endline gender weights, determined by hiring managers’ choice data. Control variables vary by specification as indicated, but the base controls (columns 2-6 and 8) include dummy variables for black and white race, male gender, college educated (or not), and continuous age and age squared. Productivity belief indicates incentivized beliefs about the relative male and female worker productivity, measured by predicted task scores; Politics refers to political party affiliation fixed effects, and “All Controls” represent all the above control variables, and additionally income and highest education level fixed effects. Columns 7 and 8 provide results for 2SLS regressions using the endline gender weight as the dependent variable, using our key independent variables as instruments for the change in beliefs about client discrimination, controlling for baseline gender weight (rather using the change in gender weight as the dependent variable).

cluding both Republican and Democrat hiring managers (see Figure C2 in the Appendix). Second, by leveraging differences in hiring choices for each manager between baseline (2 days before the election) and endline (10-12 days after the election) surveys, I remove the effects of time-invariant differences between managers across Trump and Harris states and with different baseline beliefs about the election. Given the very short period between baseline and endline surveys, it is unlikely that factors unrelated to the election affected female worker hiring over this period. Any such factors would have needed to both affect female relative to male worker hiring in the 14-17 day period between surveys and do so differently across states where Trump vs. Harris won the election. To address this unlikely threat to identification, I test for the robustness of the main results to the inclusion of different sets of control variables in regressions, such as hiring managers’ demographics, income, and political affiliation, among other controls (Table 1). If any confounding changes in female worker hiring could be predicted by managers’ observable baseline characteristics, this would account for them. The results remain nearly unchanged when including a range of different controls.¹¹

¹¹The effect of a Trump relative to Harris victory remains between 7.1 and 6 percentage points and significant in all specifications. The effect of the categorical “surprise” variable remains between 5.0 and 6.9 percentage points and

Moreover, beliefs and decision-patterns unrelated to third-party discrimination did *not* change over the election and appear unlikely to account for the observed changes in female worker hiring. First, I find that beliefs about the relative productivity of male and female workers did not change over the election (Figure C8).¹² These beliefs did not predict changes in female worker hiring, nor did they meaningfully change the main effect estimates when added to regressions (Table 1). Next, while the weight that managers placed on worker gender changed over the election, and this change was predicted by the independent variables (i.e., the winner of the election in the manager’s state, and whether the manager predicted which candidate would win), there were no such changes in the weights that managers placed on workers’ education and Quiz A scores (Appendix Figure C3). This suggests that the effects of the election on female worker hiring are unlikely to be driven by changes in beliefs about worker productivity or spurious changes in decision patterns.

2.2.3 Are Hiring Managers Accurate?

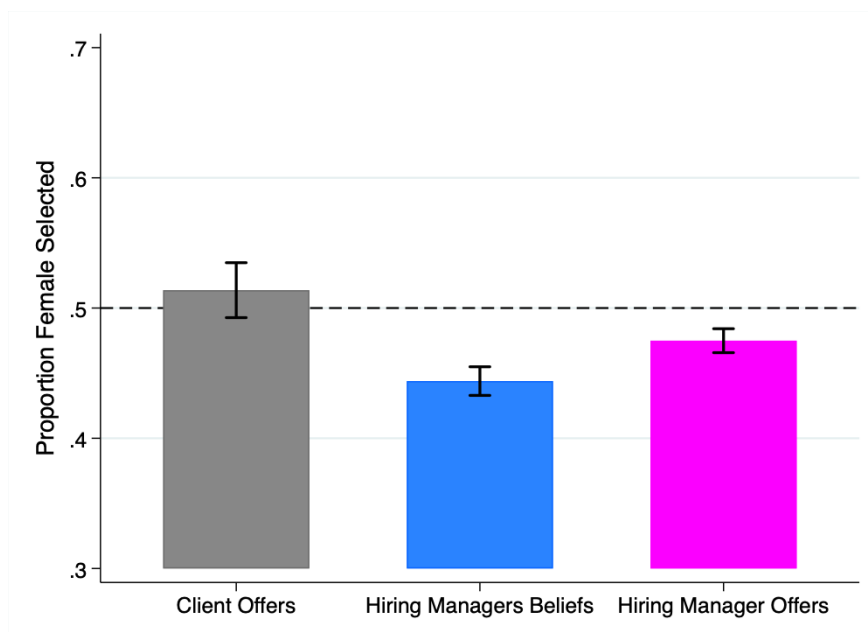
Hiring managers thus changed their hiring decisions based on their beliefs about client discrimination; but were these beliefs accurate? To test this, I compare hiring managers’ beliefs about clients’ female worker selection to clients’ actual choices, pooling across both baseline and endline surveys as pre-registered. I find that hiring managers significantly and substantially overestimated clients’ discrimination against female workers: managers believed clients would choose female workers 44.4% of the time, 6.9 percentage points less than clients’ actual female worker selection (Figure 5; one sample t-test of hiring manager beliefs equaling clients’ mean selection: $t = 5.25$, $CI = [0.044; 0.096]$, $p < 0.001$). To test whether this overestimation led hiring managers to exacerbate discrimination in this labor market, I pooled decision data for clients and hiring managers and regressed their pooled gender decision weights (the weight they placed on a worker being female) on a dummy variable for whether the decision maker was a hiring manager. This is the primary pre-registered analysis for this hypothesis. This analysis shows that hiring managers were 6.1 percentage points less likely to hire an equivalent female worker than clients were (Appendix Figure C4; $t = 4.92$, $CI = [-0.037; -0.087]$, $p < 0.001$). This effect is robust to the inclusion of different sets of control variables (Figure C4), and remarkably similar for different sub-samples of hiring managers: for all

is significant in all specifications but one, where $p = 0.059$.

¹²This analysis uses client beliefs about male and female worker productivity, as I only collect hiring managers’ endline beliefs. Clients were drawn from the same population as hiring managers.

10 tested sub-samples, point estimates are within the 95% confidence interval of the main effect and statistically less than 0 (Appendix Figure C5). This suggests that in this labor market, hiring managers discriminated against female workers more than clients did and more than they would have with accurate beliefs, demonstrating that these third-party beliefs exacerbated discrimination.

Figure 5: Comparing Hiring Manager Beliefs and Choices to Client Choices



Notes: The gray (left-most) bar represents the mean proportion of female workers clients selected. The blue (middle) bar represents hiring managers' mean belief about the proportion of female workers' clients selected. The magenta (right-most) bar represents the mean proportion of female workers hiring managers hired. In all cases, these means are for clients and hiring managers across all states and collapsed across baseline and endline choices and beliefs (as-preregistered). Error bars represent 95% confidence intervals of the means

3 Controlled Experiment

To more cleanly identify the causal effect of third-party discrimination on hiring choices, I conducted a pre-registered online experiment where I directly manipulated manager beliefs about client gender discrimination, rather than relying on naturally occurring variation in these beliefs (AsPredicted 179434). I conducted this experiment using the same hiring paradigm as in the election experiment, but included a set of treatment conditions where I varied the information provided to hiring managers and clients. Below I present the details of this design.

3.1 Design

I recruited participants from Prolific Academic who again played one of three roles in a stylized online labor market: either workers, clients, or hiring managers. For this experiment, I used the same pool of workers from the election experiment (see Section 2.1.1 for details) but recruited a new set of clients and hiring managers, in this case with no restriction on which U.S. state they lived in. The final sample included 502 participants: 300 hiring managers and 202 clients. Hiring managers and clients completed the same hiring task as in the election natural experiment, making choices between workers in each of 15 randomly assigned worker pairs where they observed each worker’s gender, highest level of education, and score on an online task (see Section 2.1.1 for more details about this task). Again, clients were incentivized to select the most productive worker in each pair and hiring managers were incentivized to select the worker that a randomly yoked client selected. As in the election experiment, hiring managers also provided incentivized beliefs about client discrimination and incentivized beliefs about both male and female worker productivity. Unlike in the election experiment, hiring managers and clients completed the hiring task and belief measures only once, and not during the election period.

3.1.1 Treatment Conditions

Before completing the hiring task, I experimentally manipulated hiring managers’ beliefs about clients’ gender discrimination by randomly assigning managers to one of three experimental conditions: the *Control* condition, the *Clients Debiased* condition, or the *Clients as-is* condition. In the *Control* condition, hiring managers completed the hiring task with no information about the true association between worker gender and productivity (i.e., Quiz B scores). Meanwhile, in both of the two treatment conditions of interest (the *Clients Debiased* condition and the *Clients as-is* condition), I showed hiring managers the true distribution of worker productivity (scores in Quiz B) by worker gender. These data showed that female workers slightly outperformed male workers, though the difference was relatively small and not statistically significant (see Figure 6). Hiring managers assigned to the *Clients Debiased* condition were told that clients would see “exactly the same” information, while managers assigned to the *Clients as-is* condition were told that clients would “not see” the same information. To ensure this was incentive compatible and deception-free,

clients were assigned to one of two experimental conditions that determined whether they observed the distribution of scores by worker gender, or not. Then, each hiring manager was yoked to a randomly selected client that either did or did not see the information about worker productivity by worker gender, depending on the hiring managers’ treatment condition (please see Figure 6 for a diagram showing all treatment conditions and each yoking between client and hiring manager conditions). Clients and hiring managers were randomly assigned to an experimental condition (and received any condition-specific information) before completing the hiring task and eliciting beliefs. Other than these manipulations, the tasks for hiring managers and clients were the same as in the election experiment.

I predicted that hiring managers would expect clients to discriminate against female workers in the absence of worker productivity information, given the gender-stereotyped nature of the task and previous evidence of discrimination against female workers on the exact same task (Bohren et al., 2022).¹³ Meanwhile, hiring managers who learned that clients saw the true relationship between productivity and worker gender should shift their beliefs about client discrimination and expect clients to become more favorable towards female workers. I verified this belief manipulation using both pilot data and data from this experiment (see Section 3.2.1 for these results). Importantly, by providing hiring managers in both treatments with the true distribution of worker scores by gender, I rule out traditional statistical discrimination (i.e., differences in beliefs about productivity by worker gender) as an explanation for differences in female worker hiring between each group. I verify that there are no differences in beliefs about worker productivity between these groups.

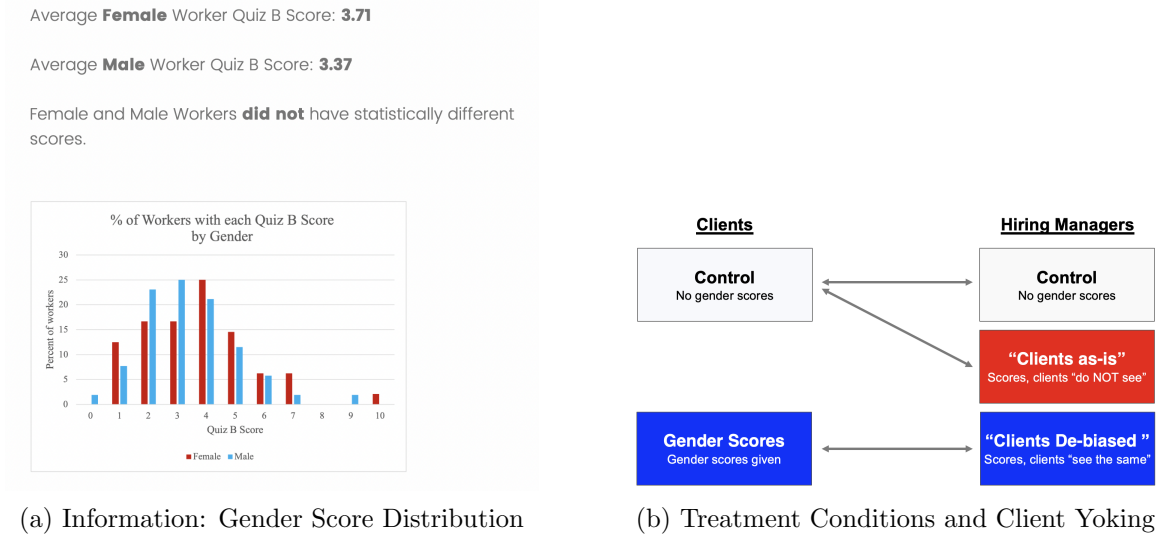
3.2 Results

3.2.1 Beliefs about Client Discrimination

First, I test whether the belief manipulation for hiring managers worked. As preregistered, I specifically focus on hiring managers’ incentivized beliefs about how many female workers clients selected

¹³There is a history of male-dominance in math, finance, history, and related tasks (Bohren et al., 2022; Bordalo, Coffman, Gennaioli, & Shleifer, 2019; Coffman, 2014; Guiso et al., 2008); female workers are also prone to self-stereotype in similar tasks, thus believing that they themselves are worse in stereotyped domains than they actually are (Coffman, 2014). Finally, previous work using the exact same task found that Prolific workers in the U.S. a) expected female workers to perform worse than male workers, and consequently discriminated against them in a hiring task, and b) believed that others also discriminated against female workers based on hiring in this task (Bohren et al., 2022).

Figure 6: Controlled Experiment: Treatment Groups and Gender Score Information



Notes: Panel (a) shows the “gender score information” provided to clients in the “Gender Score” condition and to hiring managers in the *Clients as-is* and *Clients Debiased* conditions in the Controlled Experiment. Panel (b) shows the treatment conditions for both clients and hiring managers in the Controlled Experiment and arrows represent which client treatment conditions hiring managers in each condition were randomly paired with to determine their bonuses.

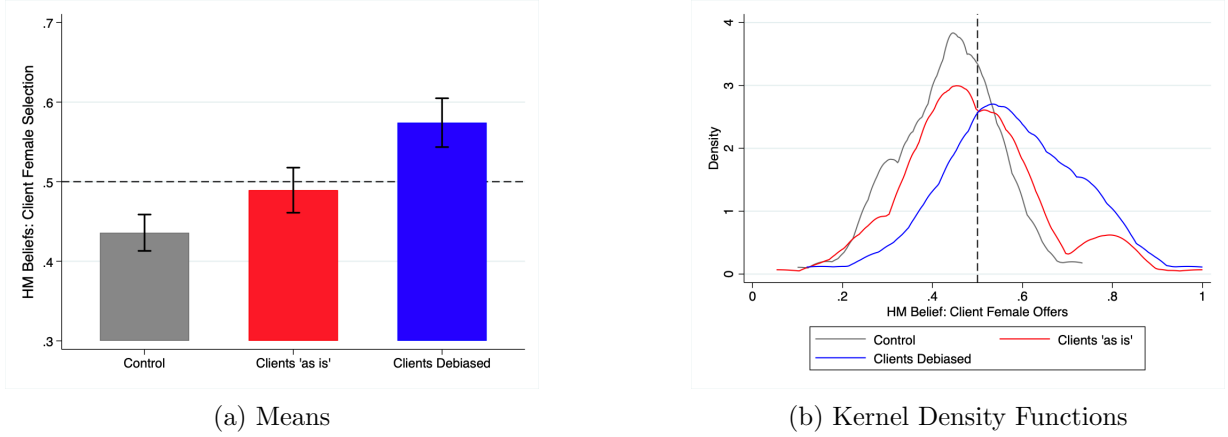
on average. I find that learning that clients received productivity information broken down by worker gender, and thus that clients were potentially “debiased,” changed hiring managers’ beliefs about client gender discrimination. Hiring managers in the *Clients Debiased* condition believed that clients hired more female workers than managers in the *Clients as-is* condition ($\beta = 0.085$, $t = 3.98$, $CI = [0.043; 0.127]$, $p < 0.001$; see Figure 7 and Table F15 in the Appendix). Surprisingly, receiving the information about worker productivity by gender was sufficient to change beliefs about clients’ female worker selection. Hiring managers in the *Clients as-is* condition believed that clients hired more female workers than hiring managers in the *Control* condition (Figure 7; $\beta = 0.053$, $t = 2.78$, $CI = [0.016; 0.091]$, $p = 0.006$).¹⁴

3.2.2 The Effect on Hiring Discrimination

Next, I examine the effect of these belief manipulations on hiring managers’ choices. Figure 8, Panel (a) shows the mean proportion of female workers hired by managers in each treatment con-

¹⁴This is surprising because in both groups, hiring managers were yoked to the same clients and had the exact same information about clients’ choices. This finding is consistent with work on the curse of knowledge, suggesting that when decision makers learn a new piece of information they are unable to suppress it when making judgments (Camerer, Loewenstein, & Weber, 1989) This is also similar to findings on “information projection”, where decision makers project their own knowledge, beliefs, or information onto others (even when others cannot possibly know this, such as in this case; Madarász (2012)).

Figure 7: Controlled Experiment: Hiring Manager Beliefs about Client Female Worker Selection by Treatment Condition



Notes: Panel a) shows hiring managers' mean beliefs of the proportion of female workers selected by clients, in each treatment condition in the controlled experiment. The dotted line represents 50% male and female workers selected by clients. Error bars represent 95% confidence intervals of the mean belief. Panel b) represents the kernel density function of these beliefs by treatment condition.

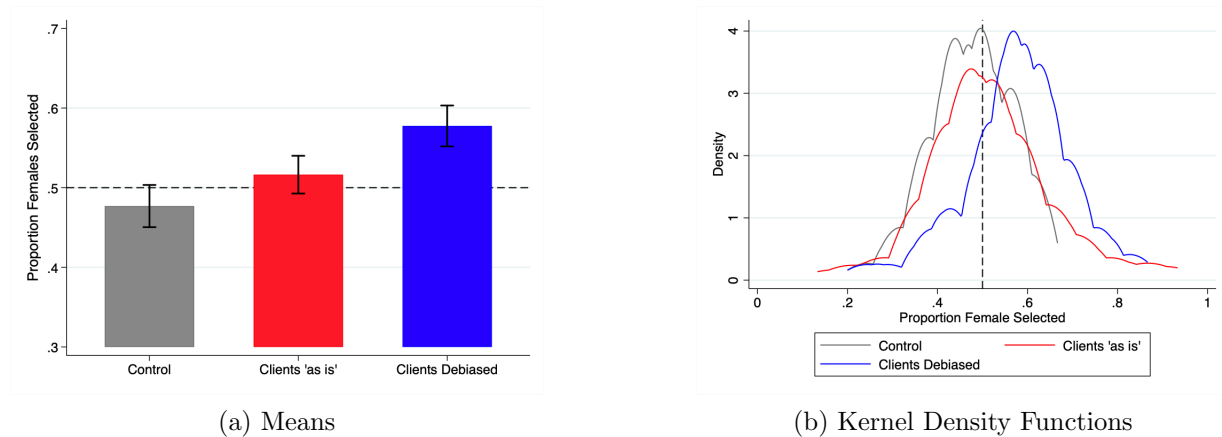
dition. Hiring managers in the *Clients Debaised* condition hired significantly more female workers than managers in the *Clients as-is* condition (Figure 8; $\Delta = 0.061$, $t = 3.43$, $CI = [0.0262; 0.0961]$, $p = 0.006$). The primary pre-specified regression analysis shows that, after controlling for workers' signals of productivity, manipulating manager beliefs about clients' choices (i.e., the *Clients Debaised* treatment) caused an 8.0% change in the likelihood that hiring managers hired a female worker (Appendix Table F15, column 4; $t = 3.33$, $CI = [0.033; 0.127]$, $p = 0.001$). This effect is robust to different specifications (Appendix Table F15). As with hiring manager beliefs, the gender score information alone had a significant effect on hiring manager choices. Hiring managers receiving this information were 6.3% more likely to choose a female worker than those in the *Control* condition (Table F15, column 4; $t = 2.96$, $CI = [0.021; 0.104]$, $p = 0.003$).

These results demonstrate that manipulating a decision maker's beliefs about third-party discrimination, in this case by clients, can affect their own choices of whether to hire a female worker. To confirm the causal channel of third-party discrimination beliefs on hiring, I again run a prespecified instrumental variable (IV) regression of hiring manager choices on their beliefs about client discrimination, using assignment to the *Clients Debaised* treatment as an instrument for these beliefs.¹⁵

¹⁵The previous subsection shows that the first-stage of this IV regression is strong: the *Clients Debaised* treatment affected beliefs about client discrimination. Given random assignment and the manipulation only differing by stating clients "did see" vs "did not see" the gender-productivity information supports the exclusion restriction for the IV is also supported.

This allows me to measure the relationship between hiring manager beliefs and hiring choices using exogenous variation in these beliefs caused by the experimental manipulation. The two-stage least squares regression results support the proposition that hiring manager beliefs about client choices causally affected their own choices, and that these choices were highly sensitive to belief changes (see Table F16 in the Appendix). For instance, a 10.0 percentage point increase in manager beliefs about the proportion of workers clients selected translated to an estimated 9.9 percentage point increase in the likelihood of hiring a female worker in a given worker pair.¹⁶

Figure 8: Controlled Experiment: Hiring Choices by Treatment Condition



Notes: Panel a) shows the mean proportion of female workers hired by hiring managers in each treatment condition in the controlled experiment, and the dotted line represents 50% male and female workers hired. Error bars represent 95% confidence intervals of the mean. Panel b) represents the kernel density function of the proportion of female workers hired by treatment condition.

I also replicate the result on hiring manager belief accuracy from the election natural experiment, finding that hiring managers significantly overestimated clients' discrimination against female workers: managers anticipated that clients would hire female workers 43.5% of the time, on average, 6.0 percentage points less than clients' actual female worker selection ($t = 5.11$, $CI = [0.036; 0.083]$, $p < 0.001$)¹⁷. Moreover, hiring managers once again exacerbated discrimination in this labor market due to their inaccurate beliefs about client discrimination: after accounting for worker signals, hiring managers were 5.4 percentage points less likely to hire an equivalent female worker than

¹⁶This was estimated using the primary pre-registered coefficient on managers' incentivized beliefs about client female worker hiring in the two-stage-least squares regressions in column 2 of Table F16: $\beta = 0.993$

¹⁷As preregistered, I specifically focus on the control groups for both hiring managers and clients (i.e., those assigned to the *Control* condition in each set of participants), since these groups' beliefs were not manipulated. I can thus test how decision makers might make these choices in the absence of (usually unavailable) information on productivity by gender, and compare to the election study where such information was not provided.

clients were (Appendix Figure E18; $t = 2.91$, $CI = [0.017; 0.081]$, $p = 0.002$)¹⁸

4 Other Analyses

4.1 What Types of Hiring Decisions were Affected?

Across 15 randomly selected worker pairs in the hiring task, there was substantial variation in workers’ relative qualifications, or signals, in each pair (i.e., differences between workers’ highest level of education and their scores on Quiz A). This allows me to test how the effect of third-party discrimination beliefs on female worker hiring differed as a function of workers’ relative signals in each pair, e.g., in pairs where male workers were more qualified than female workers vs. pairs where female workers were more qualified. I investigate heterogeneity in the effect of third-party discrimination beliefs across two dimensions of worker pairs. First, as pre-registered in the controlled experiment, I split worker pairs into categories based on the relative qualifications of workers in each pair, defining pairs where (i) one worker was better on at least one signal and no worse on the other (strictly and weakly dominated pairs), (ii) where one worker was better on one signal and worse on the other (ambiguous pairs), and (iii) where both workers had identical signals (equal pairs). This allows me to test whether changes in third-party beliefs affect female worker hiring when discrimination is more masked or *implicit*, as in ambiguous pairs, compared to where it is more transparent, as in dominated or equal pairs (Barron et al., 2024).¹⁹ I find that manipulating hiring manager beliefs about client choices had the strongest effect on choices where the pairs were *equal*: changing expected discrimination in the *Client Debaised* condition caused an increase in the likelihood of choosing a female worker of 22.8 percentage points in these pairs (Appendix Figure E21). This compares with an increase of 8.5 percentage points in ambiguous pairs and 9.8 percentage points in weakly dominated pairs. The effect for strictly dominated pairs was approximately zero.

¹⁸Here, I pool data for clients and hiring managers in the control groups and regress the choice to select a female worker in a given worker pair on a dummy for whether or not the decision maker was a hiring manager, controlling for worker signals. This constitutes the primary pre-registered analysis for this hypothesis.

¹⁹Barron et al. (2024) use the method proposed by Cunningham and De Quidt (2022) to identify hiring discrimination as both implicit (using worker pairs as in the “ambiguous” case described in this manuscript), and explicit (using worker pairs where discrimination based on gender is transparent given other signals of worker productivity). The authors find both forms of discrimination based on gender in hiring. This motivated this comparison in the present paper.

Second, I divide worker pairs by the predicted productivity differences between female and male workers in each mixed-gender pair, given their signals of productivity (i.e., each workers’ highest level of education and score on Quiz A).²⁰ I find that the *Clients Debaised* treatment had the largest effect when female and male workers were similarly qualified (i.e., they had similar predicted productivity given their qualifications). When I divide worker pairs in the controlled experiment into quintiles of relative female vs. male qualifications, I find the largest effect of the debiasing treatment in the 3rd quintile, where male and female workers were approximately equally qualified ($\beta = 0.175$, $t = 3.81$, $p < 0.001$; Appendix Figure E22). I find the smallest effects, neither of which are significant, in quintile 1 (where male workers are clearly more qualified) and quintile 5 (where female workers are clearly more qualified). In the election natural experiment, managers in Trump states compared to those in Harris states changed female worker hiring the most in quintile 4, in pairs where female workers were slightly more qualified than their male counterparts on average (Appendix Figure C11).²¹

Taken together, it appears that manipulating decision maker beliefs about third-party discrimination had a larger effect on female worker hiring when workers were more similarly qualified, and worker gender could serve as a “tiebreaker”.

4.2 The Belief-Action Gap

Figure 5 shows that hiring managers in the election experiment overestimated clients’ discrimination against female workers, and themselves hired significantly fewer female workers than clients did. However, Figure 5 also shows another pattern: that hiring managers selected significantly more female workers than they believed clients did (the gap between the blue and magenta bars). Hiring managers believed that clients selected female workers 44.3% of the time, yet themselves hired female workers 47.5% of the time ($\Delta = 0.031$, $t = 5.16$, $p < 0.001$). Hiring managers thus did not

²⁰I used a logit model to predict the worker with the highest Quiz B across all worker pairs, using their highest level of education and score on Quiz A as predictors. I then used this model to compute the likelihood that the female worker in each mixed-gender pair had a higher Quiz B score, and divided worker pairs into quintiles based on this probability.

²¹Managers in Trump and Harris states showed no difference in female worker hiring in each quintile before the election. However, after the election, their choices diverged, specifically in quintiles 3-5. That is, Trump winning in a state compared to Harris caused hiring managers to specifically hire fewer female workers in cases where female workers were similarly or more qualified than their male counterparts. This effect is largest in quintile 4, where female workers are always more qualified than male workers, but where this difference is relatively small.

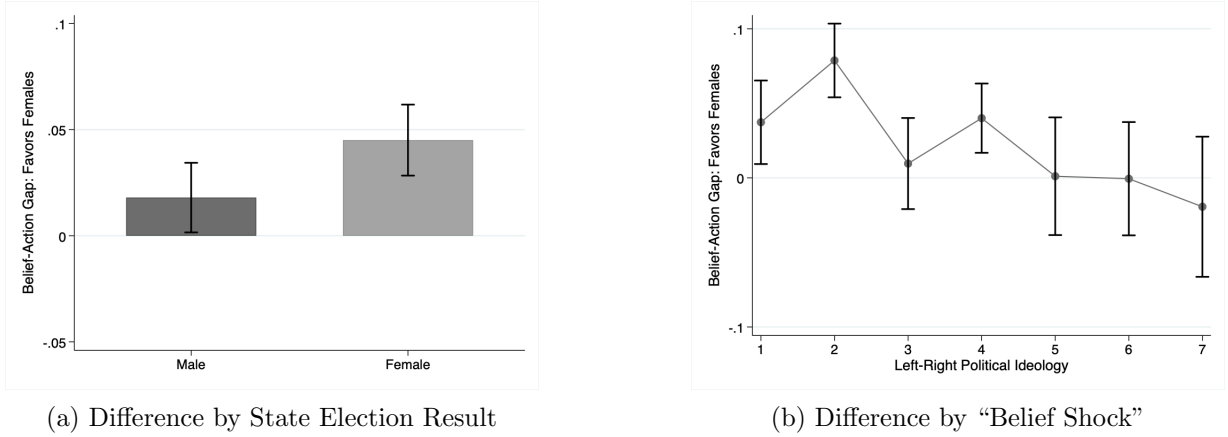
appear to fully act on their beliefs about client discrimination in their own hiring choices, even though this was the profit-maximizing strategy.²² This deviation between choices and beliefs is consistent with a “preference” to hire female workers over male workers: for a given belief about client choices, hiring managers favored female workers in choice, all-else-equal. I call this a “belief-action gap” in favor of hiring female workers.

I test several moderators of the “belief-action gap”, and while most covariates do not predict the gap, I find several significant predictors (Appendix Figure C14). First, I find that female hiring managers favor female workers in their choices, given their beliefs, by nearly 5 percentage points on average ($t = 5.28, p < 0.001$), while male hiring managers show a meaningfully and significantly lower average belief-action-gap of 1.8 percentage points ($\Delta = 0.03, t = 2.74, p = 0.006$; Figure 9). This is consistent with female managers preferring to choose women more than male managers. I also find that those on the political left and Democrats are more likely to select female workers despite their beliefs about client discrimination than are those on the political right and Republicans, respectively (Figure C14, Figure 9). This is consistent with polling data showing that those on the political left tend to favor female labor force participation more than their counterparts on the political right (Pew Research Center, 2010). I find a similar pattern of results in the controlled experiment for both manager gender and political ideology (see Appendix Figure E24). While I cannot rule out that deviations between belief estimates and choices are a function of the different elicitation methods used, robustness checks suggest that such mechanical explanations are not likely (see Appendix Figure C15), and the above moderation supports the interpretation that decision-makers with certain identities prefer to hire female workers over male workers, all-else-equal.²³

²²In the hiring task, hiring managers were only rewarded for selecting the workers that clients selected. As such, their incentivized beliefs about clients’ selection should match their own choices, since in both cases they were incentivized for accuracy.

²³One mechanical explanation for the gap between choice estimates and beliefs is that choices could be more subject to uncertainty or noise than belief estimates, which might pull choice averages towards 50% (Enke & Graeber, 2021); this might drive the gap given both means are below 50%. However, when we separate hiring managers into deciles of their beliefs about clients’ female worker selection and deciles of their average hiring choices (Figure C15) we see that in the 7th and 8th deciles average *beliefs* about client female-worker hiring are closer to 50% than average choices are. However, a similarly large belief-action gap emerges in these deciles as that for deciles 1-6, where both mean choices and beliefs were below 50%. This provides some evidence that this gap is unlikely to be driven by choice data simply being noisier and tending more towards 50%.

Figure 9: Natural Experiment: The Belief-Action Gap by Gender and Political Ideology



Notes: Panel a) shows the mean values and 95% confidence intervals for the difference between hiring managers' incentivized beliefs of the proportion of female workers clients selected, and the proportion of female workers that they themselves selected. This is shown separately for male and female hiring managers. Positive values indicate hiring more female workers than they believe clients select, and negative values indicate hiring less females than their beliefs suggest. Panel b) shows the same means and confidence intervals for hiring managers at each level of left-right political ideology.

5 Discussion and Conclusion

This paper finds that hiring managers' beliefs about clients' discrimination causally affect their own hiring choices. First, Donald Trump, as opposed to Kamala Harris, winning the highest vote share in a hiring manager's state in the 2024 U.S. presidential election increased managers' beliefs about gender discrimination by clients in their state, and in turn caused these managers to hire fewer female job candidates. Second, experimentally manipulating hiring managers' beliefs about client discrimination, holding other factors constant, significantly changed the share of female workers they hired. I also find that across both studies, hiring managers overestimated client discrimination, and consequently hired fewer female workers than they would have given accurate beliefs. Given that clients did not discriminate against female workers on average, this led to a special form of self-fulfilling prophecy: hiring managers discriminated against female job candidates exclusively because they wrongly anticipated that clients would discriminate, when in fact they did not.

The present work thus builds on research discussing the importance of discrimination by customers and other third parties in perpetuating labor market discrimination. Gary Becker (1993) said in his 1993 Nobel lecture: "Of greater significance empirically is the long-run discrimination by employees and customers, who are far more important sources of market discrimination than employers." Following Becker, empirical work shows that the demographic characteristics of an organization's

clients predict the demographics of the workers they hire (Bar & Zussman, 2017; Holzer & Ihlanfeldt, 1998; Neumark et al., 1996). Related research shows that the extent to which a position is “client-facing” also predicts gender and racial wage gaps and discrimination (Hurst et al., 2024; Kline et al., 2022). For instance, Kline et al. (2022) shows that both racial and gender discrimination were highest in sales and in retail positions with a high degree of customer interaction. However, this previous work has not experimentally tested whether beliefs about client discrimination drive these relationships, nor how these beliefs are formed. The present work directly measures and manipulates beliefs about client discrimination, testing their causal effect on hiring. The present work also adds to this existing literature by showing that public signals, such as elections, can affect these third-party beliefs.

The present paper thus links the research on customer discrimination to recent work in behavioral economics using controlled experiments in stylized labor markets to causally identify the effect of specific beliefs and belief-formation processes on discrimination (Barron et al., 2024; Bohren et al., 2023; Campos-Mercade & Mengel, 2023; Chang, Kirgios, Rai, & Milkman, 2020; Coffman et al., 2021; Esponda et al., 2023; Fischbacher, Kübler, & Stüber, 2024). This body of research has also documented the role of third-party beliefs in affecting discrimination—through a failure to account for discrimination by previous evaluators (Bohren et al., 2025, 2019) and through foreseeing discrimination by certain types of managers at later stages of the hiring process (Hirmas & Hausfeld, 2024). The present work builds on these by using a similar stylized labor market paradigm to document the influence of anticipated discrimination on final hiring decisions.

The finding in this paper that hiring managers overestimate client discrimination against female workers is consistent with related work showing that in most countries people systematically underestimate others’ support for female labor force participation (Bursztyn et al., 2023). This comports with research on “pluralistic ignorance,” showing that people assume the beliefs and actions of others differ from their own when they are in fact the same (Prentice & Miller, 1996). It is also consistent with related work on the “bias blind spot,” showing that people tend to believe others are more biased than themselves (Pronin et al., 2004, 2002). To the extent that gender discrimination is perceived as a bias, these social misperceptions might help to explain the overestimates of discrimination I observe in the present study. Similar work shows that correcting these mis-

perceptions can produce changes in female labor force participation. In an experiment in Saudi Arabia, Bursztyn, González, and Yanagizawa-Drott (2020) showed that correcting men’s beliefs about others’ support for female labor force participation (men systematically underestimated this support) increased their wives’ likelihood of being employed. This comports with the findings in the controlled experiment in the present paper, where showing hiring managers that clients are “debiased,” and thereby changing their beliefs about clients’ gender discrimination, increases the proportion of female workers that they hired.

While I explore this phenomenon with hiring managers and clients, this work could have implications for other managerial settings and for a range of third-party beliefs. For example, organizational decision-makers might anticipate discrimination by a range of different actors, e.g., a firm’s operational or team managers, existing junior employees, its shareholders, or even its suppliers. Third-party beliefs might be especially relevant for certain types of recruitment, where third parties are first-order determinants of candidate success. For example, recruitment intermediaries’ success relies on the final hiring managers’ approval of their chosen candidate. Previous work shows that hiring intermediaries, such as external recruitment agencies, are less likely to include female candidates in their shortlists (see Fernandez-Mateo & King, 2011 and Fernandez-Mateo & Fernandez, 2016), partly because they anticipate gender bias from certain types of managers, such as male or White male managers (Hirmas & Hausfeld, 2024; Lide & Berg, 2023). Hiring for executive or board of director positions might be subject to similar pressures. The hired candidate’s success in these roles relies heavily on the endorsement of shareholders, existing employees, or customers. This mechanism might help explain why women or racial minorities face particularly pronounced under-representation in these roles (Evans, 2022; Fernandez-Mateo & Fernandez, 2016; Gupta, Han, Mortal, Silveri, & Turban, 2018).²⁴

This work also has implications beyond hiring contexts. For instance, home sellers might discriminate against Black home buyers if they anticipate well-documented racial discrimination in the credit market, as these buyers might be unable to finance their home (Hurtado & Sakong, 2023; Ladd, 1998; Pope & Sydnor, 2011). Likewise, political parties or voters might disfavor prospective

²⁴Evans (2022) demonstrates that the under-representation of minority board members is largest when firms outsource board recruitment to a recruitment agency. This is consistent with the prediction that this might exacerbate the role of third-party beliefs in driving discrimination, as discussed in Fernandez-Mateo and Fernandez (2016).

female candidates because they anticipate voter discrimination against them (Corbett, Voelkel, Cooper, & Willer, 2022). Some evidence shows that voters strategically withheld votes from female candidates because they believed *others* wouldn't vote for them, making a vote for these candidates less impactful (Corbett et al., 2022). Additionally, Abraham (2020) found that entrepreneurs in male-typed occupations favored male over female contacts in a networking exchange, but only when the exchange involved a third party. It follows both theoretically and from these examples that when third-party discrimination affects decision-maker outcomes, beliefs about third parties could have upstream effects on decisions, which substantially widens the scope for the effects of these types of beliefs across economic settings. Moreover, in cases where these beliefs are overestimated, existing discrimination could be exacerbated, as observed in the studies in this paper. This could occur, for example, in situations where individual attitudes change while collective recognition of this change in society lags (Bicchieri & Mercier, 2014; Bursztyn, González, & Yanagizawa-Drott, 2020). Or when others' discriminatory behavior is highly salient and psychologically "available", such as when there is extensive media reporting about discrimination (Pronin, 2007; Pronin et al., 2002).

Several limitations of this study merit discussion. First, I cannot fully rule out the concern that participants may have responded in socially desirable ways, although the study was designed to minimize demand effects and social desirability bias. For instance, all responses were fully anonymous, and all decisions and belief estimates were incentivized, with accuracy bonuses that were large relative to participants' base pay. Reflecting this, participants were highly responsive to worker qualifications in their choices, choosing the dominant worker in a pair 95.3% of the time, irrespective of worker gender (I provide details on this and on the below points in the Appendix). Next, the hiring task was specifically constructed to reduce the salience of gender and to provide decision-makers with "cover" for selecting either male or female workers in each decision. Given that the survey was already anonymous, this should have additionally reduced participants' reputational concerns about responding in certain ways. Finally, while social desirability might pose a concern for the levels of female worker hiring in the studies, it should not necessarily drive the differences in hiring observed between treatment conditions.

Next, although the experiments in this paper were incentive-compatible and involved real hiring

processes, a key limitation is that these findings were drawn from experiments conducted online rather than in the field. While these experiments were internally valid and leveraged a real-world belief-shock, it is difficult to generalize the effect sizes observed here to what one might expect for hiring outcomes in a field setting. Similarly, these studies cannot capture potential general equilibrium effects that might exist in real-world labor markets. For instance, although I find that hiring managers discriminate against female workers due to inaccurate beliefs about client discrimination, managers may update beliefs in practice through feedback from observed outcomes—such as client behavior, peer feedback, or firm performance—leading to corrections in these beliefs over time. However, in organizational settings, such feedback is often noisy, difficult to acquire, and challenging for managers to integrate, impeding learning (Esponda, Vespa, & Yuksel, 2024; Feng & Seasholes, 2005; Fiol & Lyles, 1985; Levitt & March, 1988; Niederle & Vespa, 2023). Moreover, if managers avoid hiring women because they anticipate client discrimination, they may rarely observe client reactions to female workers. This endogenous feedback could prevent learning even when signals would otherwise be informative (Denrell, 2005; Larrick & Wu, 2007). It is thus possible that these belief-distortions persist in equilibrium (Denrell, 2005; Esponda, 2008; Niederle & Vespa, 2023). In fact, I find that managers with prior hiring experience are not more accurate in their beliefs about client discrimination than those with no experience (see Figure C5).

An important direction for future work is thus to examine how beliefs about third-party discrimination affect hiring decisions in field settings, and to monitor how these beliefs evolve over time. Field experiments or panel data could test whether beliefs and firm behavior respond to information about others’ discrimination. For example, quasi-experimental designs could use external shocks to beliefs of third-party discrimination (e.g., the Black Lives Matter protests or high-profile legal rulings, such as the U.S. 2023 ruling against affirmative action) to examine changes in third-party beliefs and contemporaneous changes in firms’ hiring behavior. Data sources like those used in Hangartner et al. (2021), which include the universe of hiring decisions and information on a large recruitment platform in Switzerland, offer the opportunity to measure changes in discrimination following similar events.

Finally, as discussed above, this framework could apply beyond hiring and gender discrimination. There is extensive scope for future research to demonstrate the role of third-party beliefs on dis-

crimination in other organizational and market contexts, to isolate where belief-biases might exist, and to test interventions to correct such biases, particularly where they exacerbate discrimination. While some evidence shows that misperceptions of third-party gender bias are widely spread (Bursztyn et al., 2023), there is scope to investigate these misperceptions more extensively, for example regarding discrimination against other groups. The present work provides early evidence that such beliefs are both consequential and potentially inaccurate, but that they are subject to change, whether by external shock or informational intervention.

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Appendix

Table of Contents

B	Natural Experiment Supplements	47
B.1	Sample Selection	47
B.2	Baseline and Endline Survey Questions	51
B.3	The Worker Pool	56
B.4	Analysis Strategy	58
B.5	Additional Analyses	61
B.6	Social Desirability Bias	64
C	Natural Experiment - Additional Figures	66
D	Natural Experiment - Additional Tables	79
E	Controlled Experiment - Additional Figures	85
F	Controlled Experiment - Additional Tables	93

B Natural Experiment Supplements

B.1 Sample Selection

I selected a set of 11 states where the election was predicted to be the more closely contested according to poll data. I started by selecting the 7 “swing states”, each of which had a predicted margin of victory lower than 2 percentage points for each respective predicted winner. In order to create more variation in states and to increase the likelihood of variation in the state-level election winner in the case of a “landslide” victory in favor of one candidate within the swing states, I selected four additional states: two that predicted Donald Trump would win by a slightly higher margin and two that predicted Kamala Harris would win by a slightly higher margin. These were Ohio and Iowa in favor of Donald Trump and Virginia and Minnesota in favor of Kamala Harris. I specifically focused on states where poll data suggested the election would be closely contested in order to increase overall uncertainty about which candidate would win, allowing the election to be more informative about which candidate was most preferred in the state (Table D6 shows that there was a high degree of uncertainty about which candidate would win within the sample). Below is a table showing each state in the sample, the predicted margin of victory, and the eventual outcome for each.

Table B1: States in sample—poll predictions and election outcomes

State	Polling Data		Election Outcomes	
	Predicted Winner	Predicted Margin	Actual Winner	Actual Margin
Ohio	Trump	6.4	Trump	11.2
Iowa	Trump	4.3	Trump	13.2
North Carolina	Trump	1.5	Trump	3.2
Arizona	Trump	1.4	Trump	5.5
Georgia	Trump	1.4	Trump	2.2
Nevada	Trump	0.7	Trump	3.1
Pennsylvania	Trump	0.1	Trump	1.8
Wisconsin	Harris	1	Trump	1.1
Michigan	Harris	1.7	Trump	1.4
Minnesota	Harris	6.3	Harris	4.2
Virginia	Harris	8.8	Harris	5.7

Notes: Poll data was taken on the 2nd of November 2024 (2 days before the 2024 U.S. presidential election) from “270towin”, a non-partisan platform that aggregates poll data across large polls predicting the U.S. 2024 election outcome.

Table B2: Final Sample Descriptive Statistics: Clients and Hiring Managers

Variable	(1) Hiring Managers		N	(2) Client	
	N	Mean/SD		Mean/SD	
Age	412	42.00 (12.24)	85	41.56 (15.14)	
Male	412	0.52 (0.50)	85	0.56 (0.50)	
White	412	0.75 (0.43)	85	0.81 (0.39)	
Black or African-American	412	0.19 (0.40)	85	0.18 (0.38)	
Other Race	412	0.07 (0.25)	85	0.06 (0.24)	
College Graduate	412	0.56 (0.50)	85	0.52 (0.50)	
Has Advanced Degree	412	0.13 (0.34)	85	0.14 (0.35)	
Imputed HH Income ('000s)	412	74.90 (47.38)	85	69.32 (47.81)	
HH Income more than 75k	412	0.41 (0.49)	85	0.36 (0.47)	
Employed Full-time	328	0.57 (0.50)	63	0.44 (0.50)	
Unemployed	328	0.23 (0.42)	63	0.32 (0.47)	
Current student	337	0.11 (0.32)	72	0.08 (0.28)	
Trump won state	412	0.84 (0.37)	85	0.95 (0.21)	
Left-Right Political Scale	412	3.51 (1.82)	85	3.52 (1.72)	
Republican Party	412	0.30 (0.46)	85	0.27 (0.45)	
Democratic Party	412	0.50 (0.50)	85	0.48 (0.50)	
Other Party/Not Disclosed	412	0.21 (0.41)	85	0.25 (0.43)	
Has Real Hiring Experience	412	0.56 (0.50)	85	0.41 (0.50)	
Speaks multiple languages	412	0.10 (0.30)	85	0.08 (0.28)	
Speaks Spanish	412	0.05 (0.22)	85	0.04 (0.19)	
Completed Endline	490	0.84 (0.37)	100	0.85 (0.36)	

Notes: This table gives mean values (and standard deviations in brackets) for each variable for the final samples of hiring managers and clients used for analysis in the natural experiment, respectively (i.e., those that completed both baseline and endline surveys).

Table B3: Final and Baseline Sample Descriptive Statistics: Hiring Managers

Variable	(1) Baseline		(2) Final Sample		T-test P-value (1)-(2)
	N	Mean/SD	N	Mean/SD	
Age	497	41.28 (12.46)	412	42.00 (12.24)	0.38
Male	497	0.51 (0.50)	412	0.52 (0.50)	0.66
White	489	0.69 (0.46)	406	0.71 (0.46)	0.57
Black or African-American	489	0.19 (0.39)	406	0.17 (0.38)	0.48
Other Race	489	0.12 (0.33)	406	0.12 (0.33)	0.98
Employed Full-time	394	0.57 (0.50)	328	0.57 (0.50)	0.91
Unemployed	394	0.22 (0.41)	328	0.23 (0.42)	0.60
Current student	407	0.13 (0.33)	337	0.11 (0.32)	0.60
Trump won state	497	0.84 (0.37)	412	0.84 (0.37)	0.97
Republican Party	497	0.29 (0.45)	412	0.29 (0.45)	0.97
Democratic Party	497	0.51 (0.50)	412	0.52 (0.50)	0.91
Other Party/Not Disclosed	497	0.20 (0.40)	412	0.20 (0.40)	0.92
Speaks multiple languages	497	0.09 (0.29)	412	0.10 (0.30)	0.80
Speaks Spanish	497	0.05 (0.22)	412	0.05 (0.22)	0.96
F-test of joint significance (P-value)					0.99
F-test, number of observations					895

Notes: This table gives mean values (and standard deviations in brackets) for each variable for both hiring managers in the final sample (i.e., completing endline and baseline surveys) and hiring managers who only completed the baseline survey. The value displayed for t-tests and the F-test are p-values. The F-test result excludes employment and student status variables due to substantial missing observations (when we include all variables, and drop observations with missing values, the p-value for the F-test is 0.99) ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

Table B4: Final and Baseline Sample Descriptive Statistics: Clients

Variable	(1) Baseline		(2) Final Sample		T-test P-value (1)-(2)
	N	Mean/SD	N	Mean/SD	
Age	100	41.16 (15.05)	85	41.60 (15.11)	0.84
Male	100	0.50 (0.50)	85	0.56 (0.50)	0.38
White	100	0.74 (0.44)	85	0.75 (0.43)	0.84
Black or African-American	100	0.13 (0.34)	85	0.13 (0.34)	0.99
Other Race	100	0.13 (0.34)	85	0.12 (0.32)	0.80
Employed Full-time	77	0.47 (0.50)	63	0.44 (0.50)	0.79
Unemployed	77	0.30 (0.46)	63	0.32 (0.47)	0.81
Current student	85	0.11 (0.31)	72	0.08 (0.28)	0.63
Trump won state	100	0.94 (0.24)	85	0.95 (0.21)	0.70
Republican Party	100	0.25 (0.44)	85	0.27 (0.45)	0.75
Democratic Party	100	0.55 (0.50)	85	0.52 (0.50)	0.66
Other Party/Not Disclosed	100	0.20 (0.40)	85	0.21 (0.41)	0.84
Speaks multiple languages	100	0.11 (0.31)	85	0.08 (0.28)	0.53
Speaks Spanish	100	0.06 (0.24)	85	0.04 (0.19)	0.44
F-test of joint significance (P-value)					0.99
F-test, number of observations					185

Notes: This table gives mean values (and standard deviations in brackets) for each variable for both clients in the final sample (i.e., completing endline and baseline surveys) and clients who only completed the baseline survey. The value displayed for t-tests and the F-test are p-values. The F-test result is using only variables without missing observations (when we include all variables, and drop observations with missing values, the p-value for the F-test is 0.99) ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

B.2 Baseline and Endline Survey Questions

B.2.1 Hiring Managers

Hiring Decisions [Baseline and Endline Surveys]:

- **[Incentivized]** Which worker do you choose? If you select the same worker the Client chose, you will receive \$[1/2] if this question is selected for payment. You will receive \$0 if you choose incorrectly. [Repeat 15 times]
 - Worker A
 - Worker B

Third-party Beliefs [Baseline and Endline Surveys]

- **[Incentivized]** Out of 15 worker pairs, how many times do you think clients from your state, [STATE], chose female workers on average? If you guess within 0.5 points of the correct average, will earn \$1 as a bonus payment if this part is selected for payment.
 - .[0–15 continuous sliding scale]

Below I will ask you some questions about your beliefs about Clients/people in your state of residence, [STATE]

- How do you think clients' (from your state) choices differed by worker gender?
 - They chose significantly more male workers
 - Slightly more male workers
 - No difference
 - Slightly more female workers
 - Significantly more female workers
- Imagine 100 randomly chosen people from your state were asked whether they agreed or disagreed with the following statement: "*Women should have the freedom to work outside of*

the home.” How many people out of 100 do you think would agree with this?

– Number that agreed [0–100 continuous sliding scale]

- How much would you say other people in your state generally agree with the following statement: ” *When jobs are scarce, men should have more right to a job than women*”?

– 1 – Strongly disagree

– 2

– 3

– 4 – Neither agree nor disagree

– 5

– 6

– 7 – Strongly agree

Productivity Beliefs [Endline Survey Only]:

- **[Incentivized]** Across all female workers, what do you think female workers’ average score was for Quiz B? If you guess within 0.5 points of the correct average, you will earn \$1 as a bonus payment if this part is selected for payment.

– .[0–10 continuous sliding scale]

- **[Incentivized]** Across all male workers, what do you think male workers’ average score was for Quiz B? If you guess within 0.5 points of the correct average, you will earn \$1 as a bonus payment if this part is selected for payment.

– .[0–10 continuous sliding scale]

State Election Beliefs [Baseline Survey Only]

- **[Incentivized]** Which candidate will win the 2024 U.S. Presidential Election in your state, [STATE]? If you guess correctly, you will earn \$1 as a bonus payment if this part is selected for payment.

- Kamala Harris
 - Donald Trump
 - Other candidate
- **[Incentivized]** By what percentage margin do you think this candidate will win in your state (ie. % more of the total votes in the state than the next candidate)? If you guess within 3 percentage points of the true margin, you will earn \$1 as a bonus payment if this part is selected for payment.
 - Percentage victory margin [0–50 continuous sliding scale]
 - How sure are you that this candidate will win in your state?
 - 1 – Very unsure
 - 2
 - 3
 - 4 – Not particularly sure
 - 5
 - 6
 - 7 – Completely sure

National Election Beliefs [Baseline Survey Only]

- **[Incentivized]** Which candidate do you think will win the 2024 US Presidential Election Nationally (ie. will win the Electoral College)? If you guess correctly, you will earn \$1 as a bonus payment if this part is selected for payment.
 - Kamala Harris
 - Donald Trump
 - Other candidate

- **[Incentivized]** By what margin of total Electoral College votes/seats do you think this candidate will win (ie. how many more votes/seats than the next candidate)? If you guess within 10 electoral seats of the true margin, you will earn \$1 as a bonus payment if this part is selected for payment.
 - Electoral votes victory margin [0–150 discrete (integer) sliding scale]
- How sure are you that this candidate will win the Electoral College?
 - 1 – Very unsure
 - 2
 - 3
 - 4 – Not particularly sure
 - 5
 - 6
 - 7 – Completely sure

B.2.2 Clients

Selection Decisions [Baseline and Endline Surveys]:

- **[Incentivized]** Which worker do you choose? If you select the worker with the highest Quiz B score, you will receive \$[1/2] if this question is selected for payment. You will receive \$0 if you choose incorrectly. [Repeat 15 times]
 - Worker A
 - Worker B

Productivity Beliefs [Baseline and Endline Surveys]:

- **[Incentivized]** Across all female workers, what do you think female workers' average score was for Quiz B? If you guess within 0.5 points of the correct average, you will earn \$1 as a bonus payment if this part is selected for payment.

– .[0–10 continuous sliding scale]

- **[Incentivized]** Across all male workers, what do you think male workers' average score was for Quiz B? If you guess within 0.5 points of the correct average, you will earn \$1 as a bonus payment if this part is selected for payment.

– .[0–10 continuous sliding scale]

B.3 The Worker Pool

100 Prolific participants completed two Tasks, which were tests of their basic math, finance, and history knowledge. Each task, Quiz A and Quiz B included 10 randomly-selected questions from these topics. After completing the tasks, workers answered several demographic questions, which included their self-reported gender and highest level of education.

To construct worker pairs for this study, workers were randomly selected from the full pool of 100 workers and paired together. I then reduced the number of pairs, over-sampling: (1) mixed-gender pairs, (2) pairs where workers were more similarly qualified according to the signals provided to decision-makers (i.e., where their Quiz A scores and highest level of education differed by less than 2 units), and (3) “ambiguous” pairs, where one candidate dominated the other on one signal (i.e., on their highest level of education or on their score on Quiz A) but not on the other (this was in order to provide decision makers with “cover” to choose the candidate they truly believe is better; see Barron et al. (2024) for details of this procedure). These strategies were predominantly used to increase statistical power to more closely mimic choices between similarly qualified job candidates, and to reduce plausible social desirability bias. The final set of worker pairs had similar characteristics to the full pool of workers (they did not significantly differ with the full worker pool on any characteristic), and male and female workers in the final pairs did not significantly differ from each other on any characteristic (see Table B5 for a summary of worker pairs). For each hiring manager and client, 15 pairs were randomly selected from the pool of worker pairs, and randomly ordered.

There were two additional exceptions to full randomization of pair selection and order, both of which were designed to reduce social desirability bias. First, the first four pairs decision makers were given were always the same, though they were randomly ordered within this set of four. These four pairs were designed to be relatively “easy decisions”, where one candidate had both a higher Quiz A score and higher education (i.e., strictly dominated the other candidate with signals other than gender). Moreover, I also over-sampled same-sex pairs within this set of four (both male and female). This was in order for decision-makers to select at least some male *and* female workers in the early stages of the survey. This would thus provide decision makers with “cover” for choosing workers they truly believed would maximize their payoffs without appearing discriminatory. The

second exception is that the 15th and final pair always included a randomly selected male-female pair where each worker had the same education level and Quiz A score. Given that gender is the only way to distinguish between these workers, it makes gender more salient, and reduces the potential cover that decision makers had. I thus placed this last so that this salience and lack of cover did not allow them to make compensating decisions afterwards.

Table B5: Stimuli Summary: Worker Characteristics by Worker Gender in Final Pairs

	Female	Male
Quiz Scores (Mean)		
Quiz A Score	3.44	3.59
Quiz B Score	3.52	3.12
Highest Education (Percentage)		
High School	0.148	0.148
Some College	0.148	0.074
College Graduate	0.444	0.593
Advanced Degree	0.259	0.185
Imputed average years of education	15.89	15.81
Pair features		
Dominant	0.500	0.500
Worker A	0.519	0.481
N	27	27

Notes: This table provides mean values for characteristics of male and female workers included in the final set of worker pairs in the hiring task (i.e., the final set of workers displayed to both clients and hiring managers during the main phase).

B.4 Analysis Strategy

For my primary analysis, I exploit both cross-state variation in which candidate won the 2024 National Election, and variation in the “surprisingness” of the election results determined by participants’ baseline beliefs about which candidate would win the election and the winning margin. I estimate the below set of specifications using ordinary least squares (OLS) regression, following my primary preregistered analyses.

$$Y_i = \beta_0 + \beta_1 TrumpState_i + \beta X_i + \epsilon_i \quad (1)$$

$$Y_i = \beta_0 + \beta_1 ElectionSurprise_i + \beta X_i + \epsilon_i \quad (2)$$

$$Y_i = \beta_0 + \beta_1 ElectionSurpriseDelta_i + \beta X_i + \epsilon_i \quad (3)$$

Y_i represents the outcome of interest for hiring manager i . In my primary pre-registered analysis, this is the *change* in the weight placed on gender by participant i when hiring workers. This measures the change in the likelihood of hiring a female worker relative to a male worker, holding other qualifications (education and aptitude signal) constant. To acquire these gender weights, I regressed each individual participants’ choices for the 15 worker pairs they observed on each worker pair’s gender difference, difference in Quiz A scores, and difference in their highest level of education. This allowed me to determine the effect of gender on each decision maker’s choices, controlling for other differences in worker qualifications.²⁵ For all participants in the final sample, I observe hiring choices both immediately before and soon after the election. I derived these “gender weights” separately for each round, and found the difference by subtracting the weight at baseline from the weight at endline. Positive values for Y_i represent an increase in the likelihood of hiring female workers after the election compared to before, holding worker qualifications constant, and negative values represent a decrease in the likelihood of hiring female workers. The first equation measures the effect of being a resident in a state where Trump won the highest vote share (the

²⁵This provided a more precise measure of gender discrimination than the proportion of female workers a participant hired, but is highly correlated with this measure ($r = 0.89$). A weight of 0 implies that the participant does not use gender to make decisions, positive values imply favoring female workers, and negative values imply favoring male workers.

variable *TrumpState*) on Y_i . Given the research described in the manuscript in Section 2.1.5, Trump winning in a given state could have affected beliefs of others’ discrimination in that state and hiring managers’ own discrimination. By using a difference variable as the primary outcome, I remove the effect of time invariant features of participants in each state, and focus on the effect of a Trump victory on changes in female worker hiring. This represents the primary “manipulation” in the study. While it is unlikely that substantive changes in beliefs or preferences relevant to gender discrimination (and unrelated to the election outcome) occurred in the 12 days between baseline and endline measurement (given its brevity and the salience of the election), we run this specification with a series of different control variables X^{26} and conduct checks described in Section 2.2.2 for robustness.

In the second equation, I test the effect of a “surprising” election outcome on changes in female worker hiring. The variable *ElectionSurprise_i* uses within-state variation in “surprisingness” of the election across hiring managers by exploiting their baseline beliefs about which candidate would win the highest vote share in their state. Trump winning in a manager’s state plausibly causes them to update their beliefs more if the manager didn’t expect this victory. The same should be true, but with a directionally opposite effect, if the manager did not expect Kamala Harris to win and she did. *ElectionSurprise_i* is coded as 1 if Trump won in participant *i*’s state and the participant predicted Harris would win, 0 if the participant correctly predicted which candidate would win, and -1 if Harris won and they predicted Trump would win. The final preregistered specification, Equation 3, makes use of variation in the continuous “margin” of surprisingness of the election outcome. This measure, reflected by the variable *ElectionSurpriseDelta_i*, captures the discrepancy between Trump’s winning (or losing) margin and the hiring manager’s prediction for Trump’s winning (losing) margin, with positive values indicating that Trump received a larger relative share of the vote than the manager expected and negative values meaning that Trump received a smaller relative share of the vote than the manager predicted. More positive values thus indicate a more surprising outcome in favor of Trump. For instance, if a hiring manager lived in Michigan, a state where Trump won the vote share by a margin of 1.4%, and predicted a

²⁶Control variables include dummy variables for black and white race, male gender, continuous variables for age, age squared, and incentivized beliefs about the relative male and female worker productivity, and fixed effects for political party affiliation, income, and highest education level.

Harris victory of 2%, this value would take 3.4, reflecting the discrepancy between their prediction and the outcome in the direction of a Trump victory. If the participant had instead predicted a Trump victory of 2%, this would take a value of -0.6.²⁷ For robustness, I compare this result to an alternative that uses the manager’s self-reported level of “certainty” in their estimate of which candidate would win (on a 7-point Likert scale) to also measure the effect of the degree of surprise of the election result in favor of Trump (these results are given in Figure B1).

I conduct the same set of analyses to measure the effect of the election on managers’ beliefs about client discrimination (for clients in their state), running the same OLS regressions using the change in managers’ incentivized beliefs about the proportion of female workers clients selected as Y_i . Conditional on this “first-stage” result, I then conduct a two-stage-least-squares (2SLS) instrumental variable regression, using the above independent variables (*TrumpState*, *ElectionSurprise*, and *ElectionSurpriseDelta*) as instruments for the change in beliefs about client discrimination, to test for the causal effect of these beliefs on hiring choices.

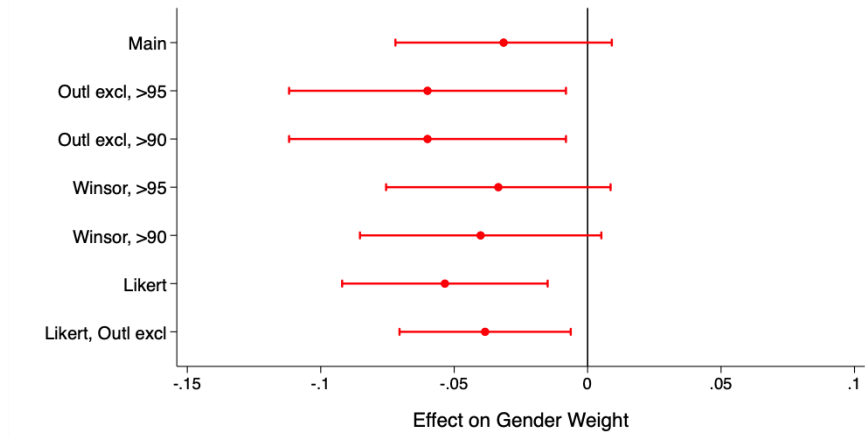
²⁷This is the key interpretation for this variable. However, in regression output, purely for presentation purposes so that coefficients are visible, I scale this variable dividing by 50 (i.e., the coefficients represent a 50% deviation between predicted and actual margin of victory)

B.5 Additional Analyses

B.5.1 Robustness of Marginal “Surprise” Effect in Election

The effect of election “surprise” on female worker hiring was only significant for a “categorical” surprise (whether or not the participant incorrectly predicted which candidate won) but not for a “marginal” surprise (the discrepancy between the expected margin of victory and the actual margin of victory), even though belief updating should theoretically be sensitive to both. To check for the robustness of the effect of the continuous measure of the elections surprisingness, I account for outliers in participants’ expectations about the margin of victory. Given the closeness of the election in the states in which I investigate, large expected margins of victory might also reflect participants being uninformed or responding inattentively (even though these measures were incentivized, and participants passed a set of attention checks). Indeed, the largest actual margin of victory across these states was 13 percentage points (pp). However, 20% of participants indicated an expected margin of victory higher than 20 pp, and 6% said that it would be higher than 40 pp. This suggests that including these participants might add noise to the analysis. I account for outliers by windsorizing and dropping outliers at different thresholds, and test an alternative measure for how “surprising” a result was (using a 7-point Likert measure of their level of confidence in their prediction of the state election winner). I find that excluding outliers for the belief of margin of victory shifts this coefficient to significance, while the alternative measure of marginal surprisingness also shows a significant negative effect (Figure B1). In spite of this significant result, however, the pattern seems to largely be driven by the “binary surprise” of which candidate won, rather than by the margin of victory. First, the coefficient on the specification with the largest effect shows that a surprise in the election outcome of 20 percentage points in favor of Trump (a very large belief shock) would lead to only a 2.5 percentage point reduction in female worker hiring, compared to a 3.5 pp reduction from failing to predict the winning candidate. Moreover, when we include the binary surprise variable in a regression with the marginal surprise variable, the effect of the marginal surprise disappears.

Figure B1: The Effect of Marginal Election Surprise on Hiring: Robustness



Notes: Red dots represent β_1 coefficients on the *ElectionSurpriseDelta_i* variable from Equation 3 in Supplement B.4, representing the effect of the deviation between managers' baseline beliefs about the election outcome and the actual election outcome. Red lines represent 95% confidence intervals of the coefficient. Each dot and line represent the coefficient for a different operationalization of *ElectionSurpriseDelta_i* using different strategies to account for outliers, which were (from top to bottom): (i) the main specification (defining *ElectionSurpriseDelta_i* as the deviation between the eventual margin of victory for the winning candidate and managers' predicted margin of victory); (ii) same definition but excluding observations where the absolute value of *ElectionSurpriseDelta_i* was greater than the 90th percentile; (iii) same definition but excluding observations where the absolute value of *ElectionSurpriseDelta_i* was greater than the 95th percentile; (iv) same definition but winsorizing at the 95th percentile; (v) same definition but winsorizing at the 90th percentile; (vi) using an alternative measure of *ElectionSurpriseDelta_i*, which is the interaction of the variable *ElectionSurprise_i* (defined in Supplement B.4; the direction of the surprise in terms of a Trump vs. Harris victory) and the managers' level of certainty (measured on a 7-point Likert scale) that their prediction of which candidate would win was correct; conceptually, this implies that the value is greater when they were incorrect *and* more certain, and thus the election should have been more surprising; (vi) the same definition excluding observations with levels of certainty of 1 or 7 (completely uncertain or completely certain).

B.5.2 Non-incentivized Belief Changes

In addition to the primary incentivized belief measure about third-party discrimination (i.e., hiring managers' estimates about the proportion of women that clients' selected), I included two measures taken from the World Values Survey and the Gallup World Poll, respectively, to elicit hiring managers' beliefs about others' more general attitudes towards female labor force participation. For beliefs about others' more general attitudes, the results are more mixed than for the main incentivized belief measure. The Gallup World poll measure, which asks managers to predict the number of people out of 100 randomly selected people from their state who agree with the statement "Women should have the freedom to work outside of the home", did not change their beliefs as predicted (Table D8, and Table D9). However, the World Values Survey measure, which is a Likert measure of how much managers think other people in their state agree with the statement "When

jobs are scarce, men should have more right to a job than women”, followed the same pattern as the incentivized measure of beliefs about client discrimination above. That is, both Trump winning a state and a surprising Trump victory in that state significantly predicted increases in expectations that others agreed jobs should be reserved for men.

B.6 Social Desirability Bias

While I cannot rule out the presence of socially desirable responding, several features of the study design and observed data limit concerns that demand effects or social desirability bias affected the main findings. First, for both experiments, all responses were anonymous and decisions and belief estimates were incentivized, with accuracy bonuses that were large relative to participants’ base pay (a \$1 or \$2 bonus for a correct randomly selected hiring choice, and 1\$ for a correct randomly selected belief estimate, compared to \$1-1.30 base pay). Reflecting this, participants were highly responsive to workers’ productivity signals in their choices. Worker signals, (i.e., their highest level of education and score on Quiz A) accounted for 31.5-34.9% of the variation in hiring managers’ choices across both experiments. Furthermore, when one worker was strictly dominant in a worker pair (i.e., had a higher level of education and a higher score on Quiz A), hiring managers chose that worker 95.3% of the time, irrespective of worker gender.

Second, the hiring task was specifically constructed to reduce the salience of gender and to provide decision-makers with “cover” for selecting either male or female workers in each decision. Given that the survey was already anonymous, this should have further reduced participants’ reputational concerns about responding in certain ways. For instance, in the natural experiment, gender was not mentioned at any point outside of worker profiles and belief elicitation questions.²⁸ Moreover, the hiring task used in both experiments included several features to limit the salience of selecting workers on the basis of gender: (i) I over-sampled worker pairs where one worker was dominant on one signal but not on the other, allowing more opportunity to justify worker selection on the basis of productivity signals (Barron et al., 2024; De Quidt, Haushofer, & Roth, 2018);²⁹ (ii) the first several worker pairs included a higher proportion of both same-sex and strictly dominant pairs, encouraging decision-makers to select male *and* female workers and to focus on productivity signals rather than gender in the early stages of the task; (iii) male-female worker pairs where both workers were identical except for gender, and thus where gender was more salient, were delayed until the

²⁸This was the same in the controlled experiment except for treatment conditions where the Quiz B score distribution broken down by worker gender was provided: importantly, none of the primary hypothesis tests in this paper rely on comparisons between participants that did vs. did not receive this information.

²⁹As discussed in Barron et al. (2024), this allows decision-makers to justify the choice of one worker over the other based on the weight they place on signals of productivity, rather than on gender. For instance, if a female worker in a pair is more educated than her male counterpart but has a lower score on Quiz A, choosing the male worker can be justified by placing more weight on Quiz A scores, claiming this to be more diagnostic of productivity.

the 15th and final pair to limit spillovers from this choice to other choices. Together, these features should have increased decision-makers “cover” for choosing workers they believed would maximize their payoffs and limiting their concerns about appearing discriminatory.

Finally, while social desirability might pose a concern for the levels of female worker hiring in the studies, it should not necessarily drive the differences in hiring observed between treatment conditions. In the controlled experiment, the main preregistered hypothesis tests involved differences in managers’ choices and beliefs between the *Clients Debiased* and the *Clients as-is* conditions. Both treatment groups received the same information about workers’ Quiz B scores by worker gender, only differing in whether they were told clients saw “exactly the same” vs. “did not see” this information. As such, gender was similarly salient for each group and socially desirable responding should not have driven differences in responses across conditions.³⁰

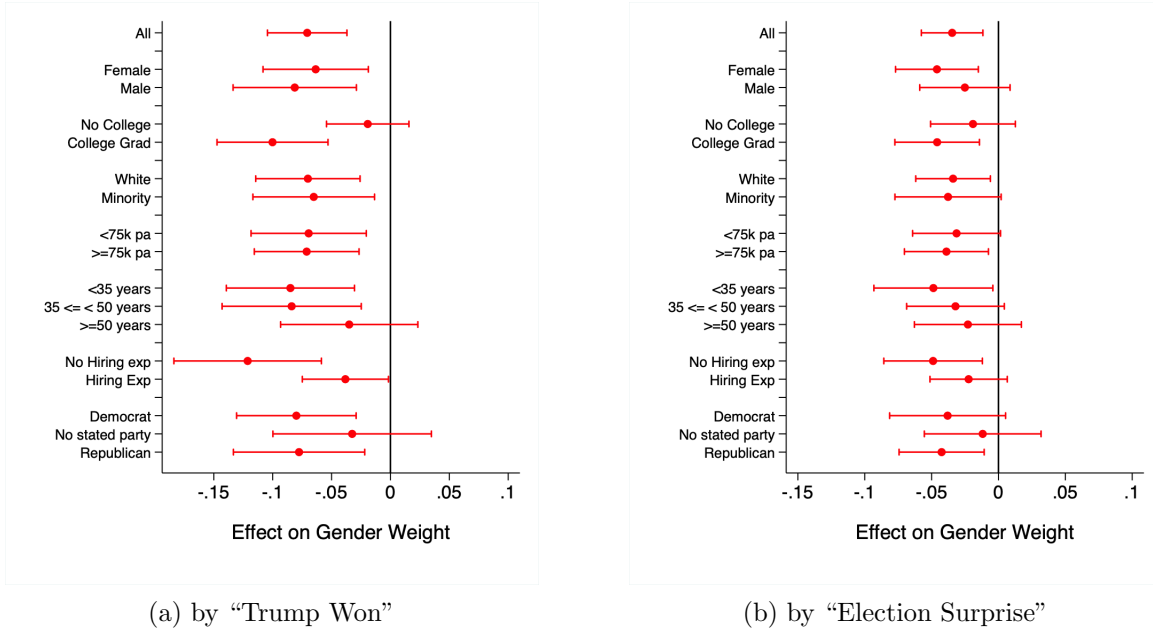
In the natural experiment, information about the experimenter and the task instructions and format were the same both before and after the election and for all managers. As such, it is unlikely that experimenter demand would have driven *changes* in manager choices over the election, particularly differentially across states and across managers with different baseline beliefs about the election.

The comparison between hiring managers and clients was potentially more vulnerable to social desirability bias. Hiring managers’ decisions and beliefs were based on potential discrimination by *others*, possibly reducing their concerns about appearing discriminatory, as they could justify their actions by appealing to others’ (rather than their own) discriminatory beliefs. Clients thus might have faced relatively more pressure to respond in socially desirable ways, as their choices reflected their own beliefs. While I cannot rule out this possibility, several features limit this concern; (i) The previously described features should limit social desirability concerns for *all* participants; (ii) hiring managers were aware of the population from which clients were selected and clients’ decision-problem, and were incentivized to accurately predict client choices: they could thus have incorporated client social desirability bias in their estimates; (iii) finally, if clients were more susceptible to social desirability than hiring managers, they should have been more biased in their choices; however, clients chose the higher performing worker more often than hiring managers did.

³⁰Gender was potentially more salient for participants in these two conditions than those assigned to the *Control* condition (who were not provided the Quiz B score distributions by worker gender). However, our primary hypothesis tests do not focus on comparisons between either condition and the *Control* condition.

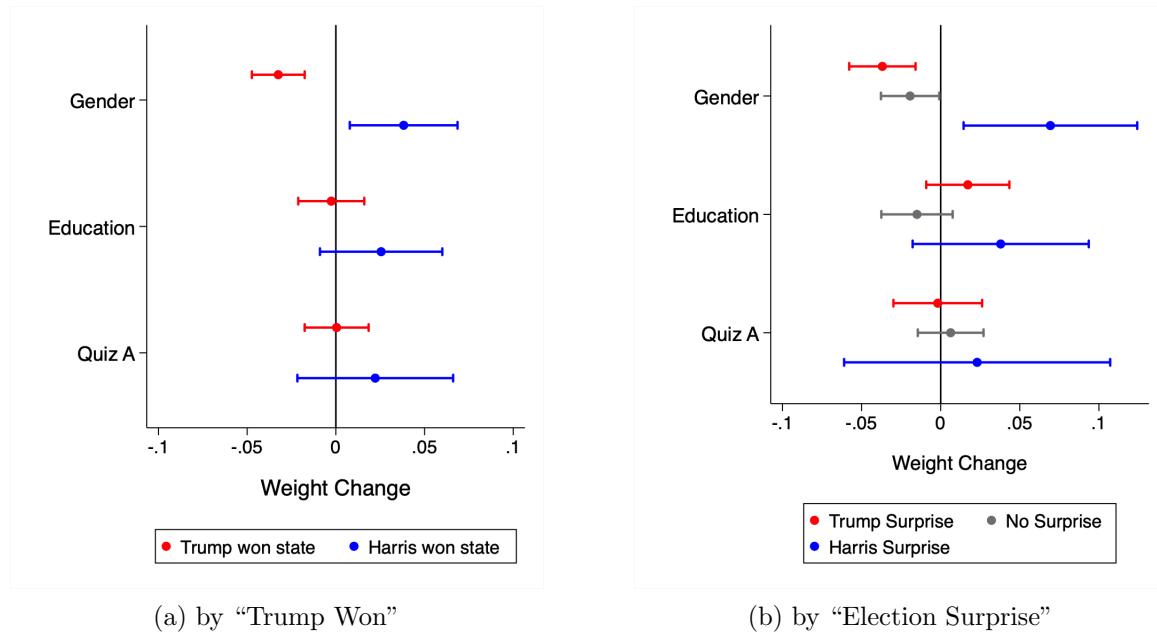
C Natural Experiment - Additional Figures

Figure C2: Change in Gender Weight in Hiring by Belief Shock: Heterogeneity



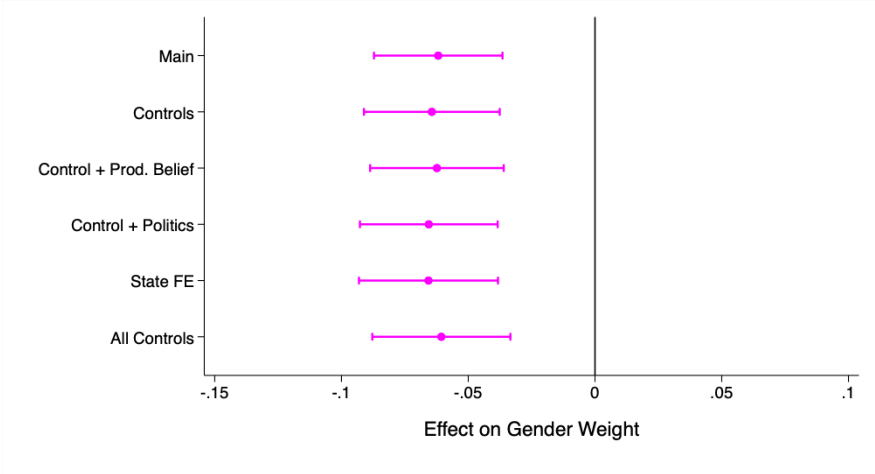
Notes: This figure shows heterogeneity in the effect of the election on changes in the decision weight placed on female workers for different “belief shocks”. In Panel (a) each dot represents the β_1 coefficient on $TrumpState_i$ from Equation 1 in the Supplement B.4 estimated separately for different subgroups of hiring managers. This tests the effect of the election outcome (i.e., being in a state where Trump vis-a-vis Harris won the election) on the change in the likelihood of hiring a female worker (represented by the weight placed on the worker being female in decisions) for different subsets of managers. The y-axis describes the specific subgroup of managers for which Equation 1 was estimated, estimating this separately based on manager gender, highest level of education, race, income, age, previous hiring experience, and political party affiliation respectively. In Panel (b) each dot represents the β_1 coefficient on $ElectionSurprise_i$ from Equation 2 in the Supplement B.4 estimated separately for the same subgroups of hiring managers. In both panels, lines represent 95% confidence intervals of the estimated coefficients.

Figure C3: Change in Decision Weights by Belief Shock



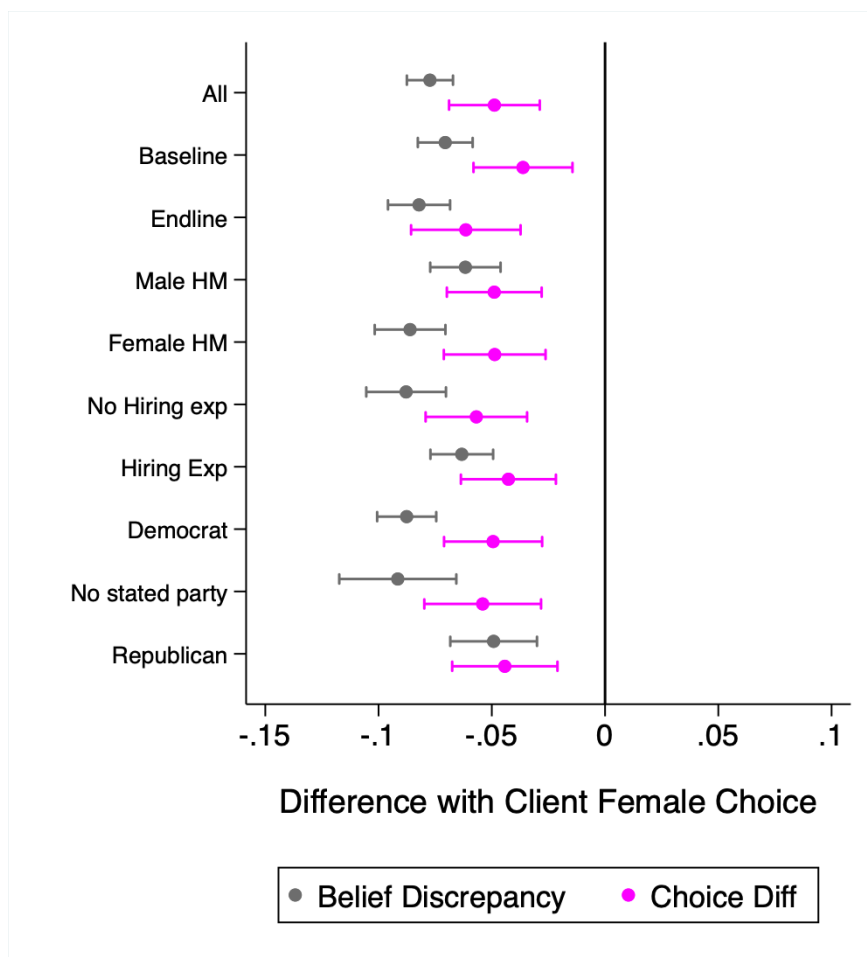
Notes: In both panels, each dot represents the mean change over the election period (i.e., between baseline and endline surveys) in the decision weights hiring manager placed on each worker attribute when selecting workers in the hiring task. In both panels, lines represent 95% confidence intervals of the mean change in decision weight. In Panel (a), this is split between managers in states where Trump won the election (the red dots) and managers in states where Harris won the election (the blue dots). In Panel (b) this is split between managers who failed to predict Trump winning in their state ("Trump Surprise"; red dots), managers who correctly predicted the winning candidate in their state ("No Surprise"; grey dots), and who failed to predict Harris winning in their state ("Harris Surprise"; blue dots).

Figure C4: Regression Estimated Difference between Hiring Manager and Client Gender Weight



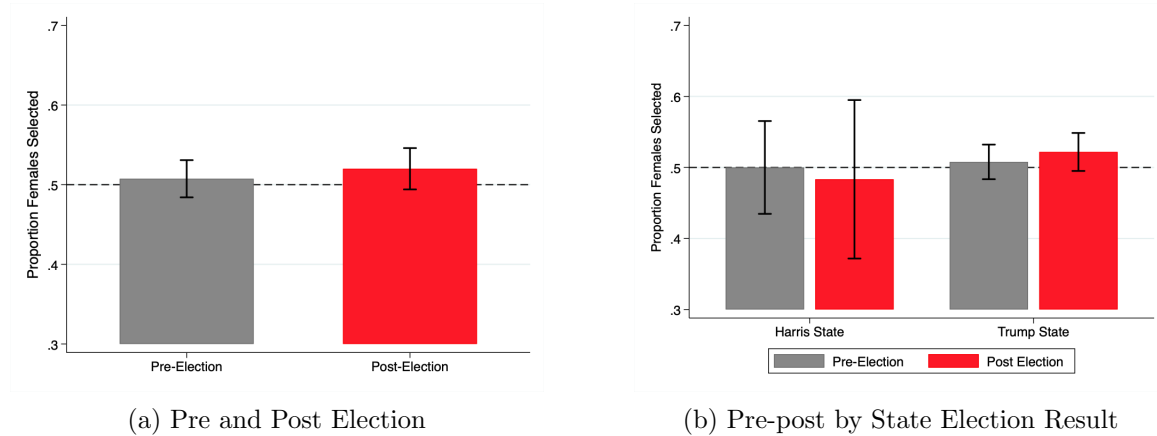
Notes: Each dot represents regression estimated difference between hiring managers and clients in the likelihood of hiring a female worker (holding other worker signals constant). The coefficient is specifically on the variable $HiringManager_i$ from an OLS regression of the decision weight placed on the worker being female (the dependent variable) on an indicator that the decision-maker was a hiring manager ($HiringManager_i = 1$) rather than a client ($HiringManager_i = 0$), and a vector of control variables X_i . This represents the primary pre-specified analysis to test the hypothesis that hiring managers and clients discriminate against female workers differently. Each row represents the results of a specification with a different set of control variables. Control variables vary by specification as indicated, but the base controls (rows 2-6) include dummy variables for black and white race, male gender, college educated (or not), and continuous age and age squared. “Prod. belief” indicates incentivized beliefs about relative male and female worker productivity, measured by predicted task scores; “Politics” refers to political party affiliation fixed effects, and “All Controls” represent all the above control variables, and additionally income and highest education level fixed effects. Lines represent 95% confidence intervals of the estimated coefficients. As pre-registered, all regressions include pooled observations (i.e., estimated decision weights for each participant) for both clients and hiring managers and at both baseline and endline.

Figure C5: Comparing both Hiring Managers' Beliefs about Client Choices and Hiring Managers' Own Choices to Clients' Choices



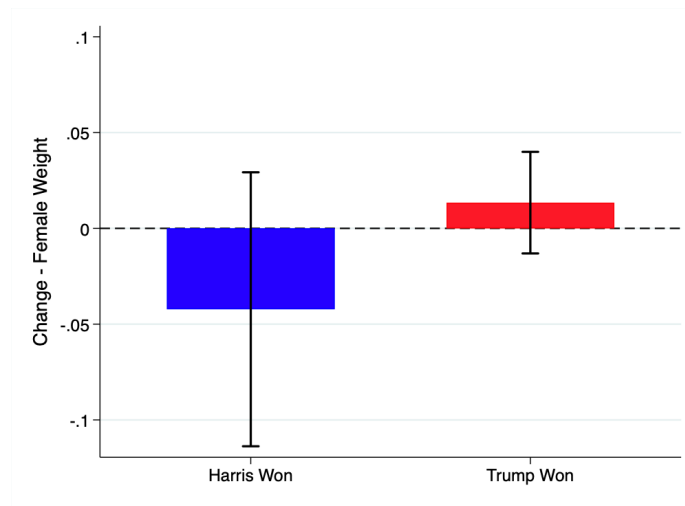
Notes: Grey dots represent mean differences between hiring managers' beliefs about the proportion of female workers that clients selected, and the proportion of female workers that clients actually selected (the "belief discrepancy"). Lines represent the 95% confidence intervals for these difference estimates. Magenta dots represent the regression estimated difference between the proportion of female workers that hiring managers selected and the proportion of female workers that clients selected (the gap between manager and client choices). Lines represent the 95% confidence intervals for these coefficient estimates. Each row represents these estimates for a different subset of the data, including broken down by baseline and endline surveys, participant gender, participants' previous hiring experience, and participants' political affiliation.

Figure C6: Clients' Selection: Proportion of Female Workers Selected



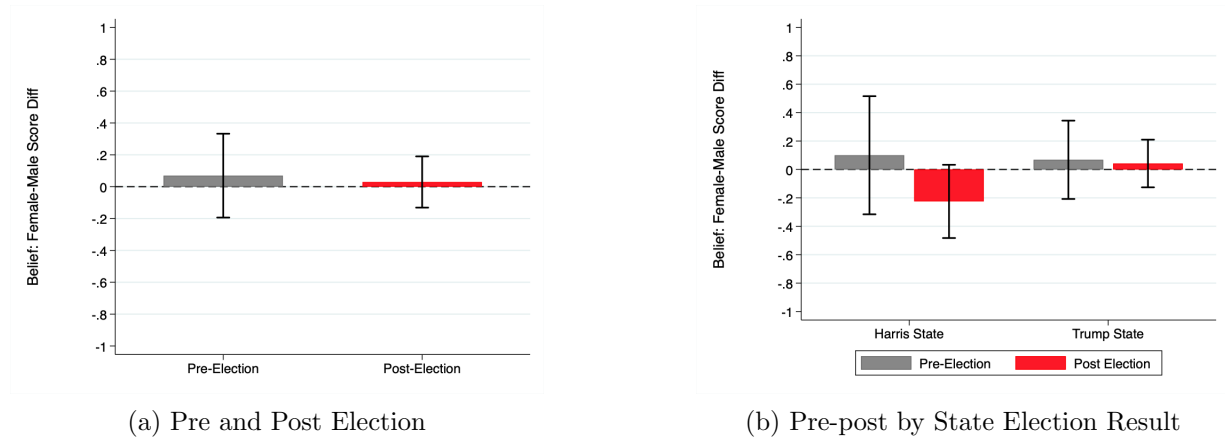
Notes: Panel a) shows the mean proportion of female workers selected by clients before and after the election, across all clients. Panel b) shows the mean proportion of female workers selected by clients, separately for i) U.S. states in which Trump won the highest share of votes in the 2024 National Election and states in which Harris won the highest share, and ii) pre-election (baseline) and post-election (endline) measures. The dotted line represents 50% male and female workers selected. Error bars represent 95% confidence intervals of the means.

Figure C7: Clients' Change in Decision Weight on Gender



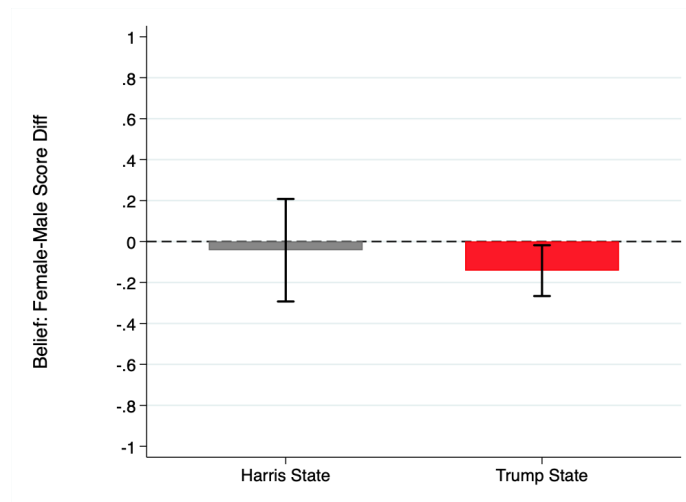
Notes: Bars show the mean change in the decision weight that client's placed on gender over the election (positive values indicate weighing female gender more positively after the election than before, and negative weights indicate weighing female gender more negatively after the election). These weight changes represent the change in the likelihood of selecting female workers vis-a-vis male workers, holding worker qualifications constant. Error bars represent 95% confidence intervals of the means.

Figure C8: Clients' Predicted Worker Scores: Female minus Male Scores



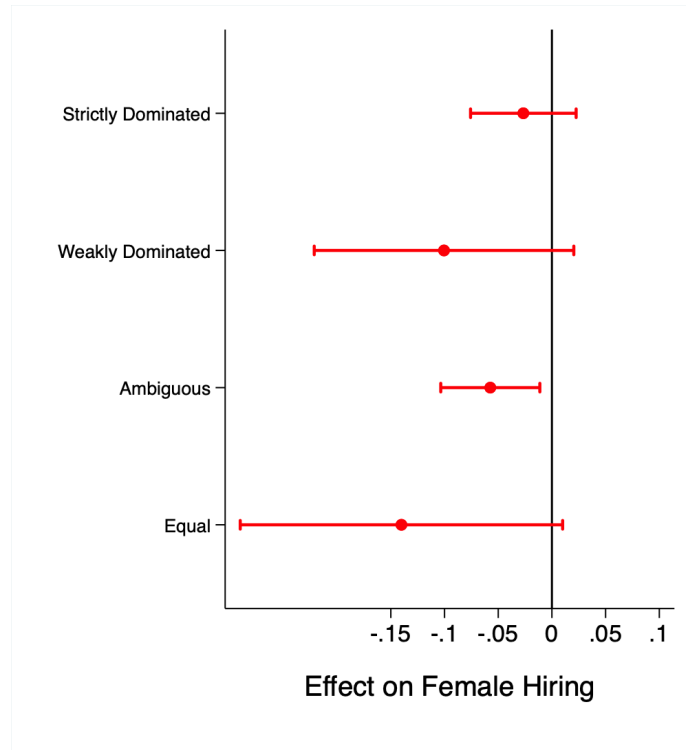
Notes: Panel a) shows the mean difference between clients' predicted Quiz B score for female workers and their predicted Quiz B score for male workers before and after the election, across all clients. Panel b) shows the same mean difference for clients, separately for i) U.S. states in which Trump won the highest share of votes in the 2024 National Election and states in which Harris won the highest share, and ii) pre-election (baseline) and post-election (endline) measures. Error bars represent 95% confidence intervals of the means.

Figure C9: Hiring Managers' Predicted Worker Scores at Endline: Female minus Male Scores, by Election Winner in State



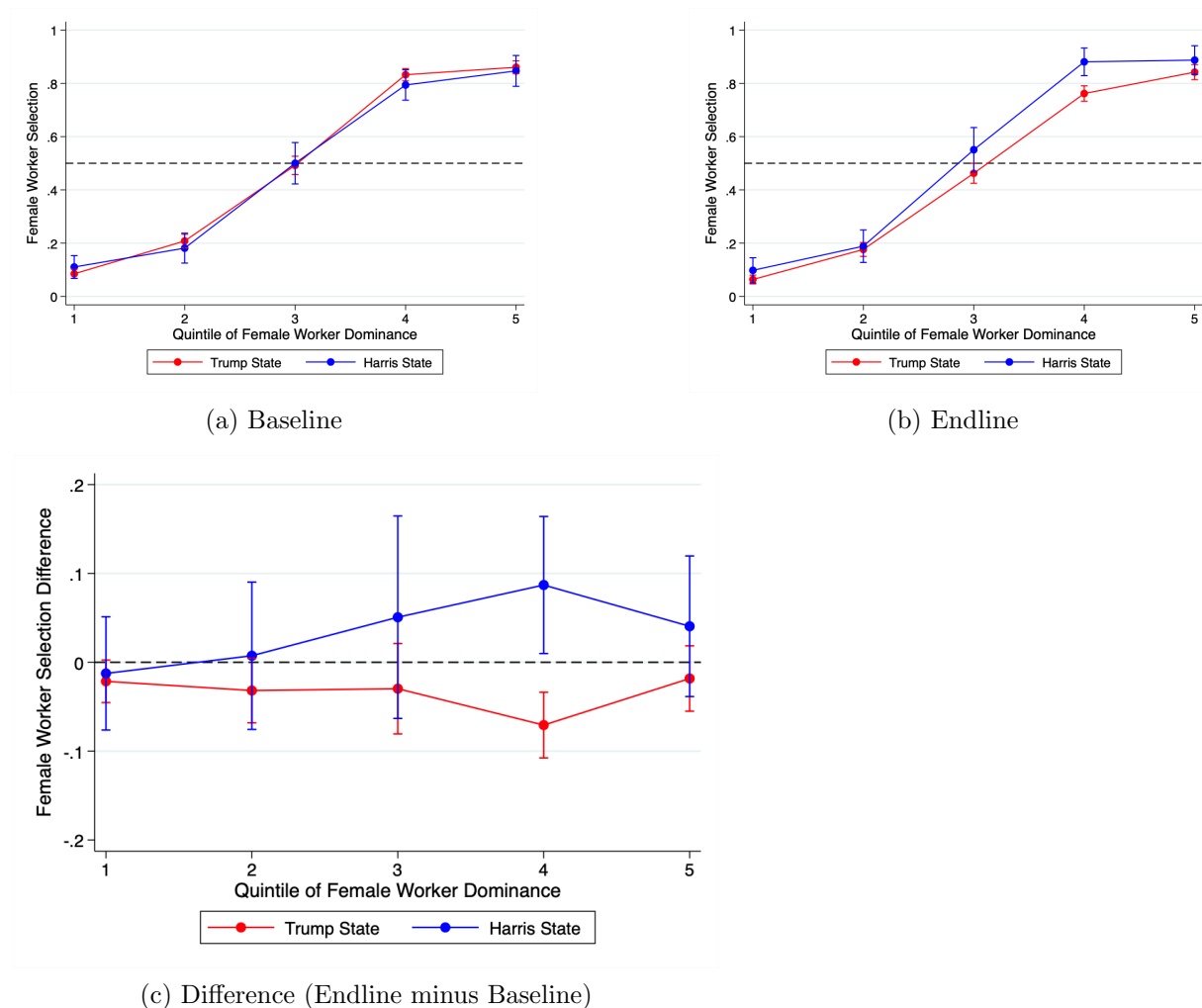
Notes: Bars show the mean difference between hiring managers' predicted Quiz B score for female workers and their predicted Quiz B score for male workers in the endline survey (post-election), separately for U.S. states in which Trump won the highest share of votes in the 2024 national election and states in which Harris won the highest share. Error bars represent 95% confidence intervals of the means.

Figure C10: The Effect of “Trump State” on Hiring, by Type of Worker Pair



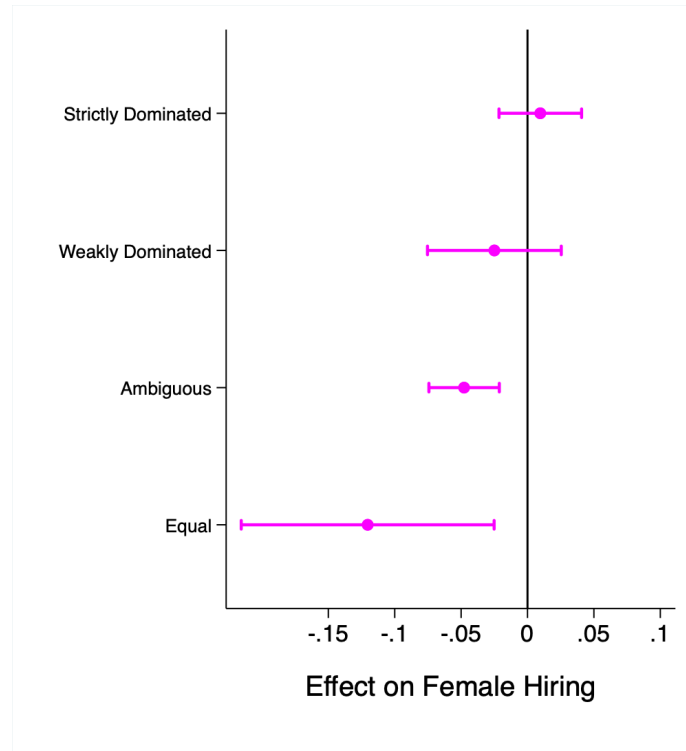
Notes: Each dot represents the β_1 coefficient on $TrumpState_i$ from Equation 1 in the Supplement B.4 estimated separately for different Y_i . Y_i represents the decision weight placed on the worker being female, holding fixed worker highest level of education and score on Quiz A (i.e., their signals) for different types of worker pairs. This tests the effect of the election outcome (i.e., being in a state where Trump vis-a-vis Harris won the election) on the change in the likelihood of hiring a female worker in worker pairs where: (i) one worker strictly dominates (based on highest education and Quiz A score), (ii) one worker weakly dominates, (iii) one worker is higher on one signal and lower on the other, and (iv) where worker signals are equal. Lines represent 95% confidence intervals of the estimated coefficients.

Figure C11: Female worker selection by state election winner,
by likelihood that female worker is more productive



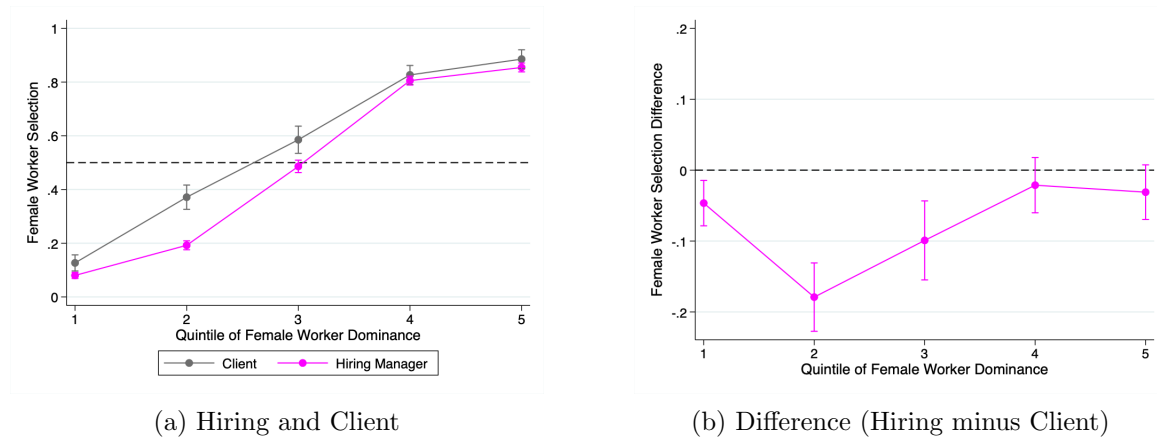
Notes: Panels (a) and (b) show the mean proportion of female workers hired by managers split (i) by whether Trump (red dots) vs. Harris (blue dots) won the highest vote share in the election in that manager's state, and (ii) across different subsets of mixed gender pairs based on the relative qualifications of female vs. male workers in each worker pair. The x-axis represents the quintile of relative female-male worker qualifications measured by the predicted likelihood that the female worker outperformed the male worker on Quiz B given their (i) highest level of education, and (ii) Quiz A score. Quintiles 1 and 2 include worker pairs where male workers were more qualified than female workers (i.e., predicted to outperform them on Quiz B), whereas quintiles 4 and 5 include worker pairs where female workers typically outperform male workers on Quiz B. Panel (a) includes results for the baseline survey and Panel (b) for the endline survey. Panel (c) shows the change in the proportion of female workers hired between baseline and endline broken down by election winner across different level of female-male worker qualification. Lines represent 95% confidence intervals of the means.

Figure C12: Hiring Manager vs Client Female Worker Selection by Type of Worker Pair



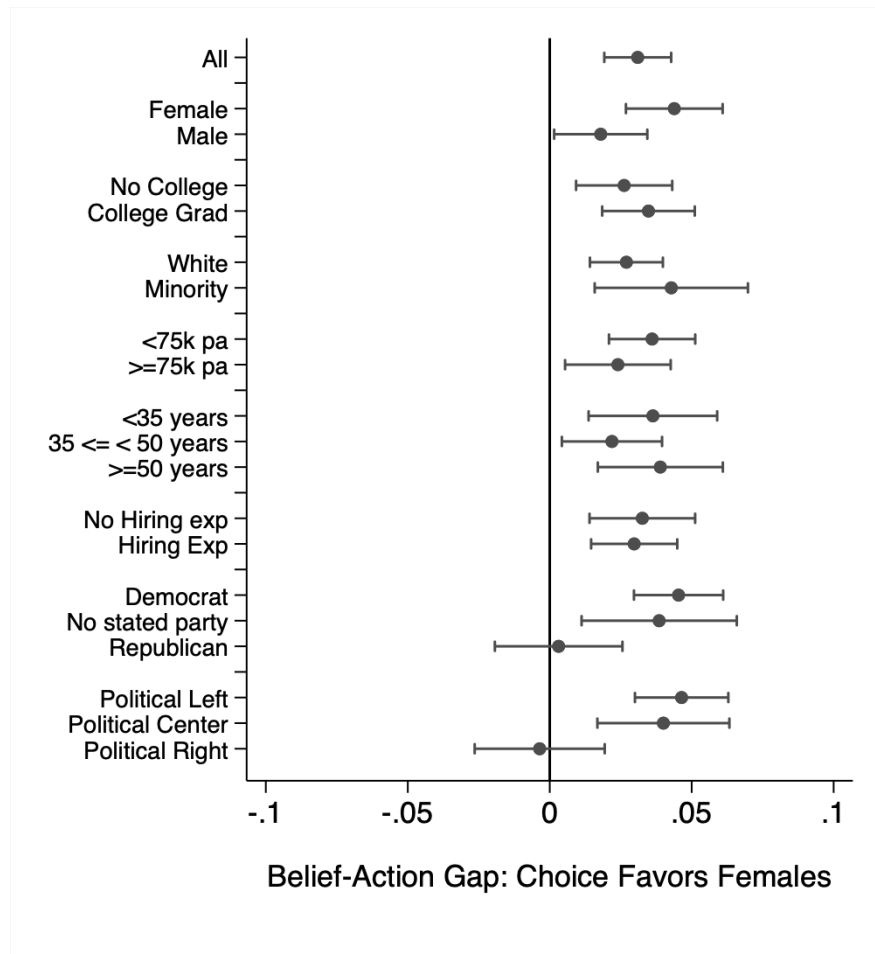
Notes: Each dot represents the regression estimated difference between hiring managers' and clients' likelihood of hiring a female worker. The coefficient is specifically on the variable $HiringManager_i$ from an OLS regression of the decision weight placed on the worker being female (Y_i , the dependent variable) on an indicator that the decision-maker was a hiring manager ($HiringManager_i = 1$) rather than a client ($HiringManager_i = 0$). I pool both client and hiring manager observations from both baseline and endline surveys when running this regression, and run it separately for Y_i derived from decisions over different types of worker pairs, such as pairs where: (i) one worker strictly dominates (based on highest education and Quiz A score), (ii) one worker weakly dominates, (iii) one worker is higher on one signal and lower on the other, and (iv) where worker signals are equal. Lines represent 95% confidence intervals of the estimated coefficients.

Figure C13: Female worker selection by Hiring Manager vs. Client,
by likelihood that female worker is more productive



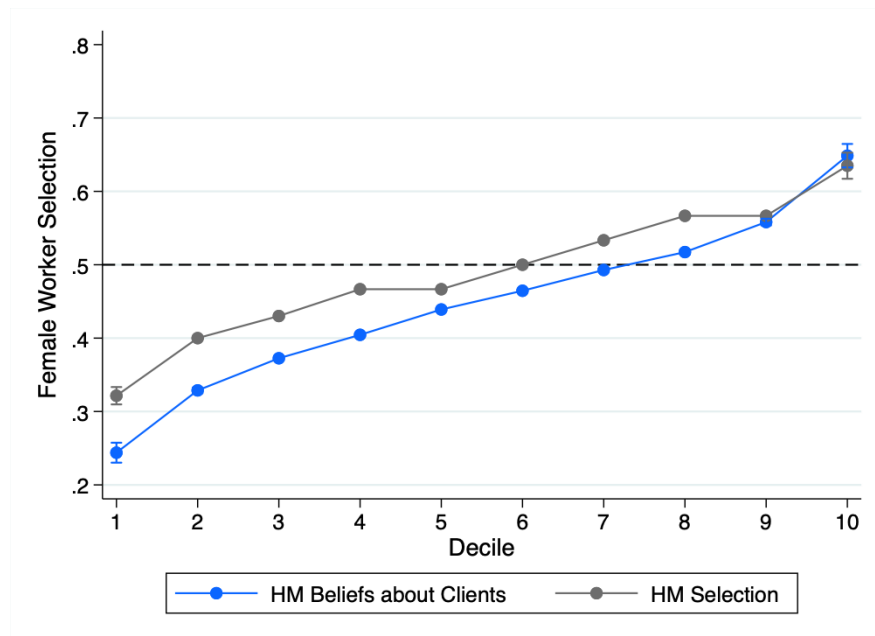
Notes: Panel (a) shows the mean proportion of female workers selected by (i) hiring managers (magenta dots dots) vs. clients (grey dots), and (ii) across different subsets of mixed gender pairs based on the relative qualifications of female vs. male workers in each worker pair. The x-axis represents the quintile of relative female-male worker qualifications measured by the predicted likelihood that the female worker outperformed the male worker on Quiz B given their (i) highest level of education, and (ii) Quiz A score. Quintiles 1 and 2 include worker pairs where male workers were more qualified than female workers (i.e., predicted to outperform them on Quiz B), whereas quintiles 4 and 5 include worker pairs where female workers typically outperform male workers on Quiz B. Panel (b) shows the different in the proportion of female workers selected between clients and hiring managers across different level of female-male worker qualification. Lines represent 95% confidence intervals of the means.

Figure C14: Belief-Action Gap Means: Heterogeneity



Notes: Each dot shows the mean difference between hiring managers' beliefs about the proportion of female workers' that clients selected and the proportion of female workers that hiring managers' themselves selected (i.e., the "belief-action gap") for different subsets of hiring managers. The y-axis describes the specific subgroup of managers for which this gap was estimated, estimating this separately based on manager gender, highest level of education, race, income, age, previous hiring experience, political party affiliation, and political ideology respectively. Lines represent 95% confidence intervals of the mean difference in each subgroup.

Figure C15: Hiring Manager Beliefs about Clients' Choices and Own Choices by Decile of Each



Notes: The grey line and dots represent the mean proportion of female workers hired by hiring managers within each decile of the same variable. The blue line and dots represent the hiring managers' mean belief about the proportion of female workers that clients selected within each decile of that variable. Lines represent 95% confidence intervals of the mean within each decile of each.

D Natural Experiment - Additional Tables

Table D6: Hiring Manager Baseline Beliefs about Election Results

	State-level				National-level			
	All States	Harris-Likely States	Swing States	Trump-Likely States	All States	Harris-Likely States	Swing States	Trump-Likely States
Belief - which candidate would win (proportion)								
Harris victory	0.52	0.63	0.57	0.13	0.59	0.52	0.61	0.55
Trump Victory	0.47	0.38	0.43	0.85	0.40	0.46	0.38	0.43
Other candidate Victory	0.00	0.00	0.00	0.01	0.01	0.01	0.01	0.01
Margin of victory beliefs (% Vote Difference)								
Absolute	12.74	17.23	11.65	13.09	7.35	9.31	7.26	5.47
Net, in Trump's favor	-0.48	-2.53	-1.80	8.98	-1.52	0.35	-2.21	-0.09
Conditional on Trump Victory Belief	12.85	19.60	11.53	12.76	7.17	10.34	6.53	6.04
Conditional on Harris Victory Belief	12.65	15.80	11.74	15.22	7.48	8.38	7.73	5.00
Certainty of candidate choice (Likert scale, 1-7)								
All	4.90	5.07	4.69	5.36	4.66	4.84	4.64	4.53
Conditional on Trump Victory Belief	4.93	5.38	4.83	4.67	4.67	4.60	4.76	4.24
Conditional on Harris Victory Belief	4.92	5.26	4.77	5.27	4.67	4.71	4.71	4.37

Notes: This table summarizes hiring managers' beliefs about the election outcome at both the national-level and the state-level. These were measured during the baseline survey (i.e., before the election). Beliefs include (i) which candidate would win (incentivized choice), (ii) the predicted winning margin of this candidate (given in percentages here; incentivized), (iii) their level of self-reported certainty that their prediction about the winning candidate was correct (from 1-completely unsure, to 7- completely certain; unincentivized).

Table D7: Regression Results for Hiring Manager Belief Changes
(of Client Discrimination) by “Belief Shock”

	1	2	3	4	5	6	7	8
Trump won state	-0.062*** (0.022)	-0.059*** (0.022)	-0.059*** (0.022)	-0.057*** (0.022)		-0.059*** (0.022)	-0.061*** (0.018)	-0.061*** (0.018)
Surprise: election winner	-0.051*** (0.014)	-0.05*** (0.014)	-0.05*** (0.014)	-0.045*** (0.014)	-0.049*** (0.018)	-0.045*** (0.015)	-0.051*** (0.011)	-0.049*** (0.011)
Surprise: winning margin	-0.134*** (0.05)	-0.138*** (0.051)	-0.138*** (0.051)	-0.121** (0.051)	-0.138*** (0.053)	-0.118** (0.051)	-0.122*** (0.04)	-0.122*** (0.041)
	Main	Control	Control + Productivity belief	Control + Politics	Control + State Fixed Effects	All controls	Baseline Belief	Baseline Belief + Controls

Notes: Each column and row correspond to a separate OLS regression specification, with coefficients on the primary independent variable reported and robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Each row represents regressions with a different set of independent variables, representing the β_1 coefficient in regression equations 1, 2, and 3 respectively (shown in the Supplement B.4). The dependent variable (Y_i in columns 1-6) is the change in hiring manager's incentivized beliefs about client discrimination, calculated by subtracting the baseline from the endline belief measure (of beliefs of the proportion of workers clients selected that are female). Control variables vary by specification as indicated, but the base controls (columns 2-6 and 8) include dummy variables for black and white race, male gender, college educated (or not), and continuous age and age squared. Productivity belief indicates incentivized beliefs about the relative male and female worker productivity, measured by predicted task scores; Politics refers to political party affiliation fixed effects, and “All Controls” represent all the above control variables, and additionally income and highest education level fixed effects. Columns 7 and 8 provide results for regressions using the alternative specification recommended by (McKenzie, 2012), which regresses endline beliefs on our key independent variables, controlling for baseline beliefs (rather using the change in beliefs as the dependent variable).

Table D8: Hiring Manager Third-Party Discrimination Beliefs:
Client Hiring and General Others in the State,
Baseline and Endline Means by State Winner and Prediction

	Round	Trump State		Harris State	
		Trump Sur- prise	Trump Ex- pected	Harris Ex- pected	Harris Sur- prise
Client Female Hiring	Baseline	0.448*** (0.01)	0.453*** (0.011)	0.461* (0.022)	0.43** (0.033)
	Endline	0.405*** (0.01)	0.452*** (0.011)	0.481 (0.02)	0.5 (0.032)
Client Female Hiring Likert	Baseline	2.575*** (0.062)	2.576*** (0.062)	2.725** (0.124)	3.192 (0.184)
	Endline	2.31*** (0.062)	2.541*** (0.065)	2.675*** (0.126)	3.038 (0.162)
Gallup out of 100 "work out home"	Baseline	79.966 (1.402)	84.343 (1.076)	83.125 (2.929)	78.731 (3.706)
	Endline	78.971 (1.181)	81.448 (1.366)	80.600 (3.128)	80.962 (3.179)
WVS Likert: "Jobs for men"	Baseline	4.678 (0.13)	4.680 (0.138)	4.575 (0.299)	4.462 (0.377)
	Endline	4.19 (0.129)	4.715 (0.132)	4.875 (0.254)	4.962 (0.326)
Observations		174	172	40	26

Notes:. Columns 2-5 display mean levels of hiring manager beliefs about third-party discrimination (both discrimination by clients and general others in their state) at both baseline and endline. Standard errors in parentheses. This is split by both states in which Trump won (columns 2 and 3) vs. Harris won the highest vote share (columns 4 and 5) and by the "surprisingness" of the result given participants' baseline prediction about which candidate would win the election ("Expected" indicates they correctly predicted the winner and "Surprised" that they failed to predict the winner). Stars represent statistical significance at the following thresholds *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$ - for tests against benchmarks of neither favoring nor disfavoring female workers (only for Client Female Hiring and Client Female Hiring Likert, as there are no normative benchmarks for the other measures). These are 0.5 for Client Female Hiring, 3 for Client Female Hiring Likert. Higher values indicating more positive attitudes towards female workers.

Table D9: Hiring Manager Third-Party Discrimination Beliefs: Change (over Election) by State Winner and Prediction: Client Hiring and General Others in the State

	Trump State			Harris State		
	Trump Sur- prise	Trump Ex- pected	Diff: Surprise - Expected	Harris Ex- pected	Harris Surprise	Diff: Surprise - Expected
Change in Client Female Hiring	-0.043*** (0.011)	-0.001 (0.014)	-0.041** (0.018)	0.021 (0.021)	0.07* (0.04)	0.05 (0.042)
Change in Client Female Hiring Likert	-0.264*** (0.07)	-0.035 (0.065)	-0.229** (0.096)	-0.05 (0.147)	-0.154 (0.233)	-0.104 (0.262)
Change in Gallup out of 100 "work out home"	-0.994 (1.435)	-2.895** (1.318)	1.901 (1.949)	-2.525 (2.801)	2.231 (3.369)	4.756 (4.41)
Change in WVS Likert: "Jobs for men"	-0.489*** (0.141)	0.035 (0.133)	-0.523*** (0.194)	0.3 (0.218)	0.5 (0.339)	0.2 (0.384)
Observations	174	172		40	26	

Notes: Columns 1-2 and 4-5 display mean changes over the election period (i.e., between baseline and endline surveys) in hiring manager beliefs about third-party discrimination. Standard errors in parentheses. This is split by both states in which Trump won (columns 1 and 2) vs. Harris won the highest vote share (columns 4 and 5) and by the "surprisingness" of the result given participants' baseline prediction about which candidate would win the election ("Expected" indicates they correctly predicted the winner and "Surprised" that they failed to predict the winner). Stars represent statistical significance at the following thresholds *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$ - for tests against 0, which represents no change, with higher values indicating more positive beliefs about attitudes towards female workers after the election. Columns 3 and 6 display the difference between values in columns 1 and 2 and between values in columns 5 and 4, respectively (i.e., "Surprise" columns minus "Expected" columns). Again, stars represent statistical significance at the following thresholds *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$ - for tests against 0.

Table D10: Hiring Manager Female Worker Selection: Baseline and Endline Means by State Winner and Prediction

	Round	Trump State		Harris State	
		Trump Sur- prise	Trump Ex- pected	Harris Ex- pected	Harris Sur- prise
Proportion of Female Workers Hired	Baseline	0.494 (0.009)	0.484* (0.009)	0.475 (0.017)	0.479 (0.022)
	Endline	0.458*** (0.009)	0.45*** (0.009)	0.493 (0.015)	0.526 (0.025)
Gender (Female) Weight	Baseline	0.008 (0.009)	-0.004 (0.009)	-0.006 (0.017)	-0.018 (0.021)
	Endline	-0.028*** (0.01)	-0.032*** (0.011)	0.012 (0.017)	0.051** (0.024)
Education Weight	Baseline	0.088*** (0.012)	0.108*** (0.012)	0.072*** (0.023)	0.034* (0.02)
	Endline	0.105*** (0.012)	0.086*** (0.011)	0.089*** (0.025)	0.072*** (0.024)
Quiz A Weight	Baseline	0.307*** (0.012)	0.317*** (0.011)	0.315*** (0.027)	0.297*** (0.04)
	Endline	0.305*** (0.012)	0.32*** (0.011)	0.337*** (0.018)	0.32*** (0.032)
Observations		174	172	40	26

Notes:. Columns 2-5 display mean levels of hiring manager female worker selection, including the proportion hired and the respective regression-estimated decision weights, at both baseline and endline. Standard errors in parentheses. Decision weights are estimated by regressing choices for each participant on the difference in gender, education signal, and Quiz A signal for each worker pair. These statistics are split by both states in which Trump won (columns 2 and 3) vs. Harris won the highest vote share (columns 4 and 5) and by the “surprisingness” of the result given participants’ baseline prediction about which candidate would win the election (“Expected” indicates they correctly predicted the winner and “Surprised” that they failed to predict the winner). Stars represent statistical significance at the following thresholds *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$ - for tests against a benchmark of 0.5 for the Proportion of Female Workers Hired and against 0 for all regression-estimated decision weights.

Table D11: Hiring Manager Female Worker Selection: Change (over Election) by State Winner and Prediction

	Trump State			-	Harris State		
	Trump Sur- prise	Trump Ex- pected	Diff: Surprise Expected		Harris Ex- pected	Harris Surprise	Diff: Surprise - Expected
Proportion of Female Workers Hired	-0.036*** (0.011)	-0.034*** (0.011)	-0.002 (0.015)		0.018 (0.019)	0.046 (0.03)	0.028 (0.034)
Gender (Female) Weight	-0.037*** (0.011)	-0.028*** (0.011)	-0.009 (0.015)		0.018 (0.018)	0.069** (0.028)	0.051 (0.031)
Education Weight	0.017 (0.013)	-0.023* (0.013)	0.04* (0.021)		0.017 (0.023)	0.038 (0.028)	0.02 (0.036)
Quiz A Weight	-0.002 (0.014)	0.003 (0.012)	-0.005 (0.018)		0.022 (0.025)	0.023 (0.043)	0.001 (0.046)
Observations	174	172			40	26	

Notes: Columns 1-2 and 4-5 display mean changes over the election period (i.e., between baseline and endline surveys) in hiring manager female worker selection, including changes in the proportion hired and changes in the respective regression-estimated decision weights. Decision weights in each period are estimated by regressing choices for each participant on the difference in gender, education signal, and Quiz A signal for each worker pair. Standard errors of mean differences in parentheses. These statistics are split by both states in which Trump won (columns 1 and 2) vs. Harris won the highest vote share (columns 4 and 5) and by the “surprisingness” of the result given participants’ baseline prediction about which candidate would win the election (“Expected” indicates they correctly predicted the winner and “Surprised” that they failed to predict the winner). Stars represent statistical significance at the following thresholds *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$ - for tests against 0, which represents no change, with higher values indicating increases in these proportions or weights after the election. Columns 3 and 6 display the difference between values in columns 1 and 2 and between values in columns 5 and 4, respectively (i.e., “Surprise” columns minus “Expected” columns). Again, stars represent statistical significance at the following thresholds *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$ - for tests against 0.

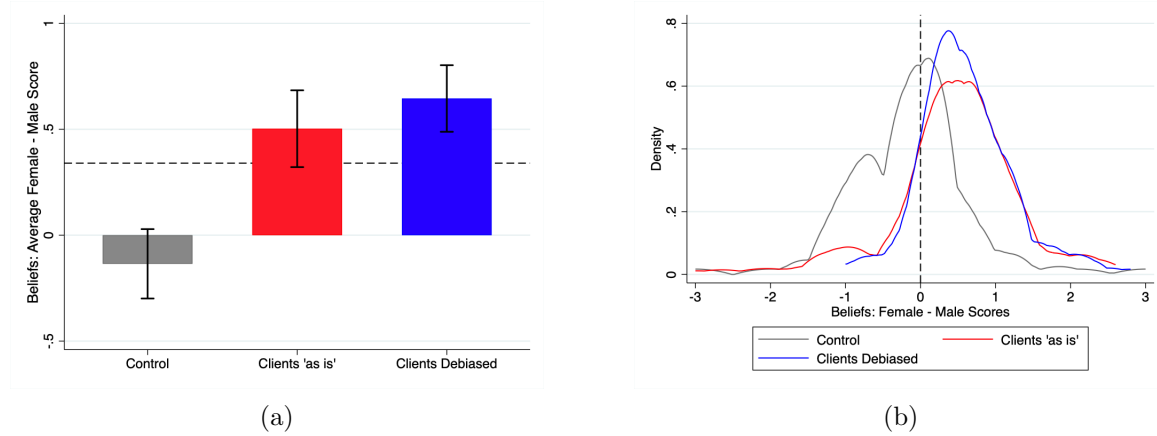
Table D12: Regression Results for Client Choices and Beliefs, Regressed on State Election Winner

	Gender Weight in Choice					
	1	2	3	4	5	6
Trump won state	0.056 (0.062)	0.045 (0.063)	0.046 (0.058)	0.045 (0.059)	0.048 (0.057)	0.033 (0.058)
	Beliefs about Female-Male Productivity					
	1	2	3	4	5	6
Trump won state	0.299 (0.727)	0.248 (0.734)	0.207 (0.728)	0.171 (0.732)	0.268 (0.39)	0.23 (0.367)
	Main	Control	Control Politics	+ All controls	Baseline Con- trol	Baseline Be- lief + Con- trols

Notes: Each column and row correspond to a separate OLS regression specification, with coefficients on the primary independent variable (an indicator for whether Trump won the highest vote share in the clients' state of residence) reported and robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. In the top panel, the dependent variable (in columns 1-4) is the change in the weight the client placed on gender (the weight placed on the worker being female) when making choices from before to after the election, calculated by subtracting the baseline from the endline gender weights, determined by clients' choice data. In the bottom panel, the dependent variable (in columns 1-4) is the change in clients' beliefs about female minus male scores on Quiz B (i.e., the change, over the election, in the predicted gender difference in actual productivity). This is calculated by the subtracting the baseline gender difference belief from the endline gender difference belief. Control variables vary by specification as indicated, but the base controls (columns 2-6) include dummy variables for black and white race, male gender, college educated (or not), and continuous age and age squared. Politics refers to political party affiliation fixed effects, and "All Controls" represent all the above control variables, and additionally income and highest education level fixed effects. Columns 5 and 6 provide results for regressions using the alternative specification recommended by (McKenzie, 2012), which regresses endline gender weight (top panel) and endline belief of gender-score differences (bottom panel) on the Trump State indicator, controlling for the baseline measure of each dependent variable, respectively (rather than using the *change* variables as the dependent variables).

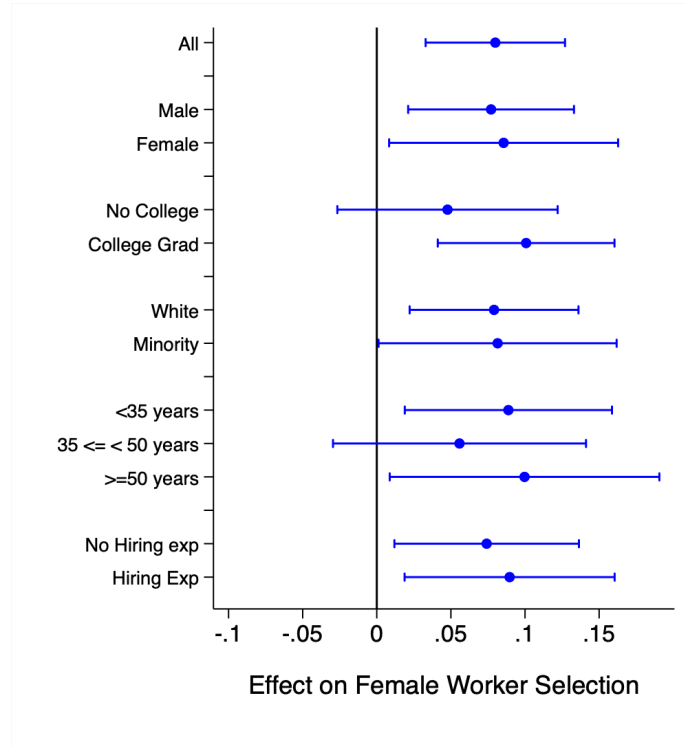
E Controlled Experiment - Additional Figures

Figure E16: Controlled Experiment: Hiring Manager Beliefs about Female-Male Worker Productivity, by Treatment Condition



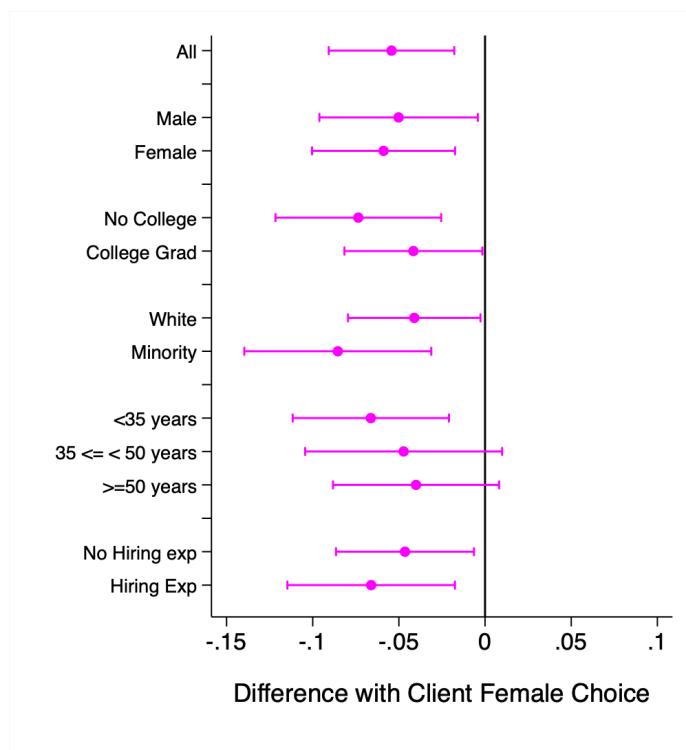
Notes: Panel a) shows the mean difference between hiring managers' predicted Quiz B score for female workers and their predicted Quiz B score for male workers, in each treatment condition in the controlled experiment. The dotted line represents the actual difference (female workers scored on average 0.336 points more than male workers). Error bars represent 95% confidence intervals of the mean belief difference. Panel b) represents the kernel density function of these belief differences by treatment condition.

Figure E17: Controlled Experiment: Heterogeneity of Effect of *Clients Debiased*



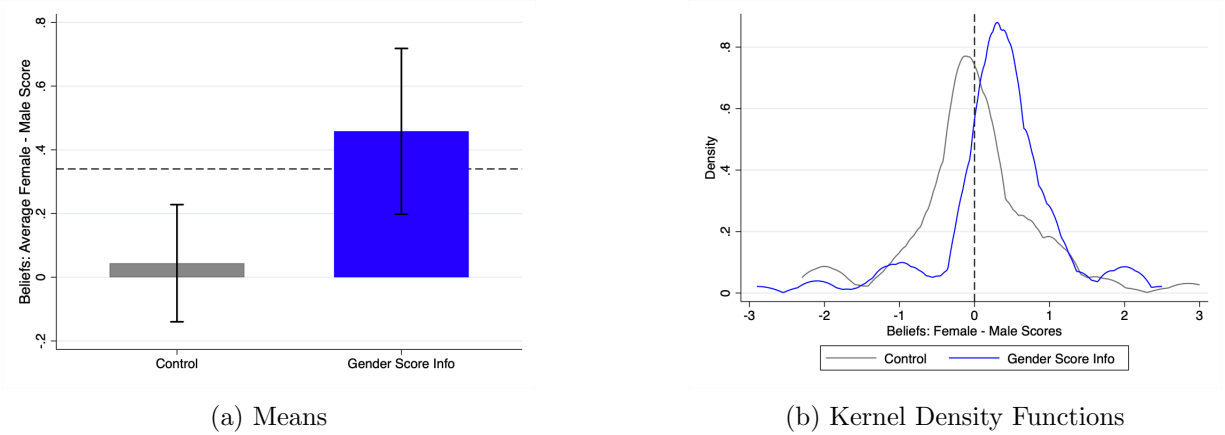
Notes: Each dot represents the coefficient on $ClientsDebiased_i$ from the following pre-registered regression model (estimated using OLS), for different subgroups of hiring managers: regressing a binary indicator for whether or not a hiring manager selected a female worker in a given worker pair on an indicator of whether the hiring manager was assigned to *either* the *Clients Debiased* condition or the *Clients as-is* condition ($GenderScoreInformation_i$; i.e., that they saw the worker gender score distribution), an indicator for whether the hiring manager was assigned to the *Clients Debiased* condition $ClientsDebiased_i$, and controlling for workers' relative scores on Quiz A and highest level of education, with standard errors clustered by participant. The y-axis describes the specific subgroup of managers for which this regression was estimated, estimating this separately based on manager gender, highest level of education, race, age, and previous hiring experience respectively.

Figure E18: Controlled Experiment: Heterogeneity in Wedge between Hiring Manager and Client Female Worker Selection



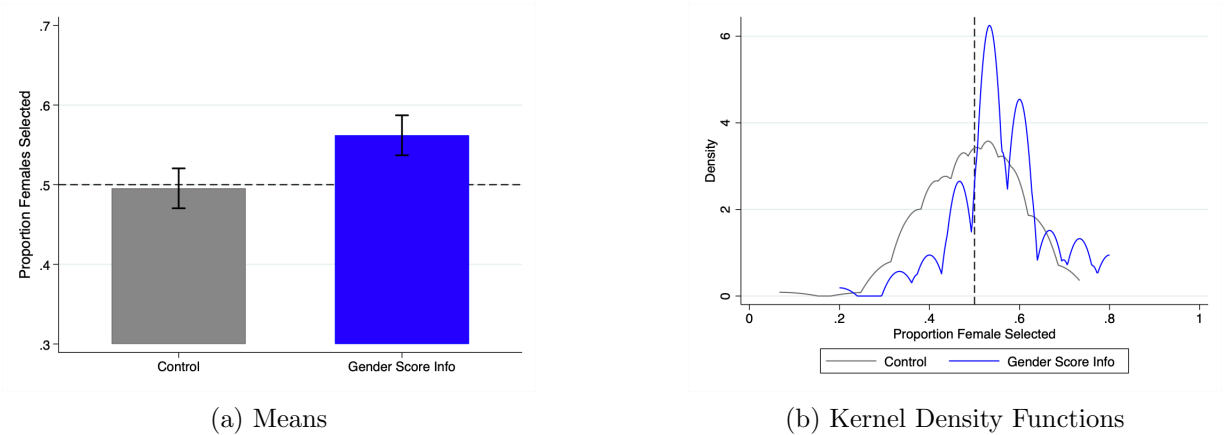
Notes: Each dot represents the coefficient on $HiringManager_i$ from the following pre-registered regression model (estimated using OLS), for different subgroups of participants (i.e., hiring managers and clients): we pool observations (i.e., pair-level decisions in the hiring task) for both hiring managers and clients assigned to the *Control* condition, and regress a binary indicator for whether or not a participant selected a female worker in a given worker pair on an indicator of whether the participant was a hiring manager (as opposed to a client; $HiringManager_i$), and controlling for workers' relative scores on Quiz A and highest level of education, with standard errors clustered by participant. The y-axis describes the specific subgroup of managers for which this regression was estimated, estimating this separately based on manager gender, highest level of education, race, age, and previous hiring experience respectively.

Figure E19: Controlled Experiment: Client Beliefs about Female-Male Worker Productivity by Treatment Condition



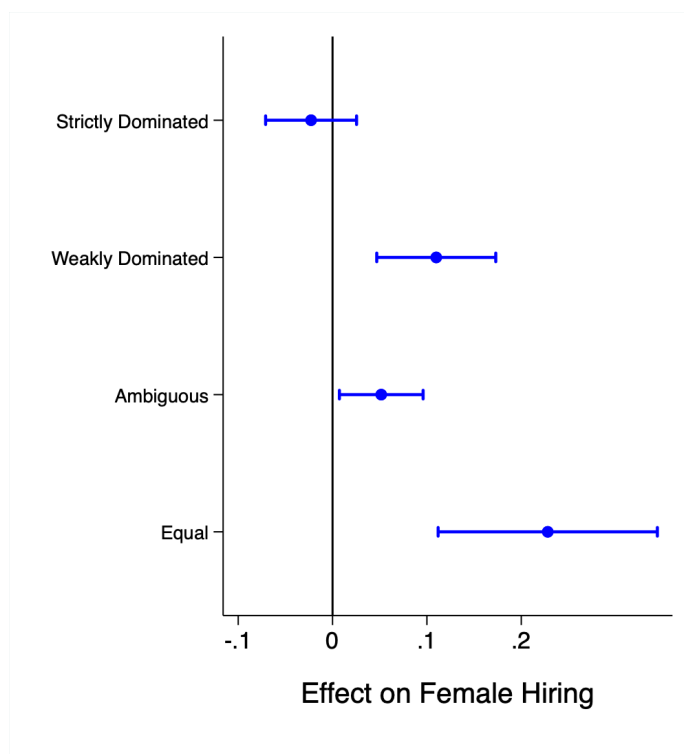
Notes: Panel a) shows the mean difference in clients' predicted Quiz B score for male workers minus the predicted Quiz B score for female workers, in each treatment condition in the controlled experiment. The dotted line represents the true difference of 0.337. Error bars represent 95% confidence intervals of the mean belief difference. Panel b) represents the kernel density function of these prediction differences by treatment condition.

Figure E20: Controlled Experiment: Client Female Worker Selection by Treatment Condition



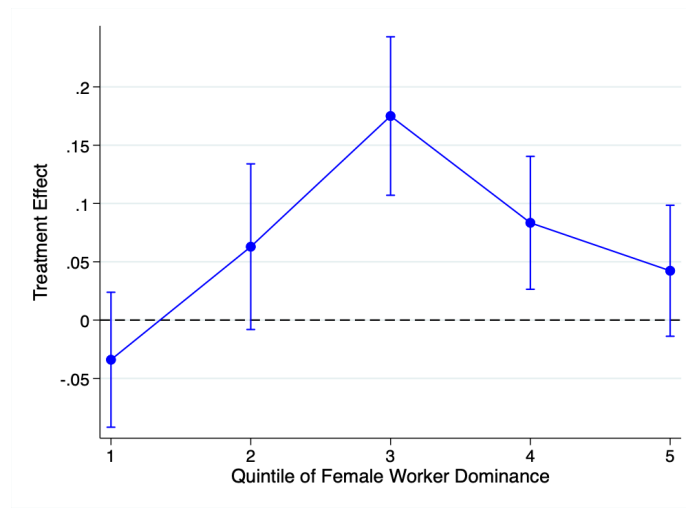
Notes: Panel a) shows the mean proportion of female workers clients selected, in each treatment condition in the controlled experiment. The dotted line represents the true difference. Error bars represent 95% confidence intervals of the mean belief. Panel b) represents the kernel density function of these choices by treatment condition.

Figure E21: Controlled Experiment: Effect of *Clients Debiased* by Type of Worker Pair



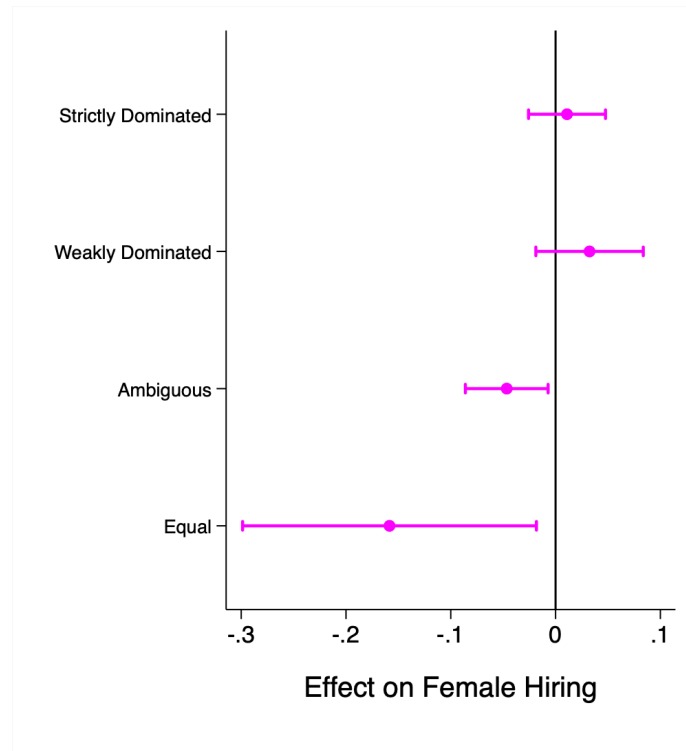
Notes: Each dot represents the coefficient on $ClientsDebiased_i$ from the following pre-registered regression model (estimated using OLS), run on different sets of worker pairs from the hiring task: regressing a binary indicator for whether or not a hiring manager selected a female worker in a given worker pair on an indicator of whether the hiring manager was assigned to *either* the *Clients Debiased* condition or the *Clients as-is* condition ($GenderScoreInformation_i$; i.e., that they saw the worker gender score distribution), an indicator for whether the hiring manager was assigned to the *Clients Debiased* condition $ClientsDebiased_i$, and controlling for workers' relative scores on Quiz A and highest level of education, with standard errors clustered by participant. This regression is estimated separately for different worker pairs, where: (i) one worker strictly dominates (based on highest education and Quiz A score), (ii) one worker weakly dominates, (iii) one worker is higher on one signal and lower on the other, and (iv) where worker signals are equal. Lines represent 95% confidence intervals of the estimated coefficients.

Figure E22: Controlled Experiment: The Effect of Client-Debiased Treatment by likelihood that female worker is more productive



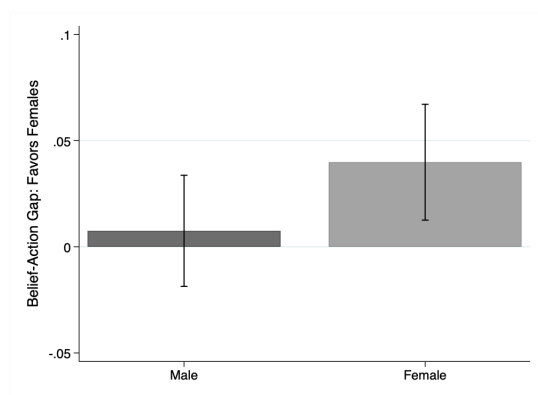
Notes: Blue dots show the coefficient on $ClientsDebiased_i$ from the following pre-registered regression model (estimated using OLS), run on different sets of worker pairs from the hiring task: regressing a binary indicator for whether or not a hiring manager selected a female worker in a given worker pair on an indicator of whether the hiring manager was assigned to *either* the *Clients Debiased* condition or the *Clients as-is* condition ($GenderScoreInformation_i$; i.e., that they saw the worker gender score distribution), an indicator for whether the hiring manager was assigned to the *Clients Debiased* condition $ClientsDebiased_i$, and controlling for workers' relative scores on Quiz A and highest level of education, with standard errors clustered by participant. I run this regression separately across different subsets of mixed gender pairs based on the relative qualifications of female vs. male workers in each worker pair. The x-axis represents the quintile of relative female-male worker qualifications measured by the predicted likelihood that the female worker outperformed the male worker on Quiz B given their (i) highest level of education, and (ii) Quiz A score. Quintiles 1 and 2 include worker pairs where male workers were more qualified than female workers (i.e., predicted to outperform them on Quiz B), whereas quintiles 4 and 5 include worker pairs where female workers typically outperform male workers on Quiz B. Lines represent 95% confidence intervals of the estimated coefficients in each quintile.

Figure E23: Controlled Experiment: Hiring Manager vs. Client Female Worker Selection by Type of Worker Pair

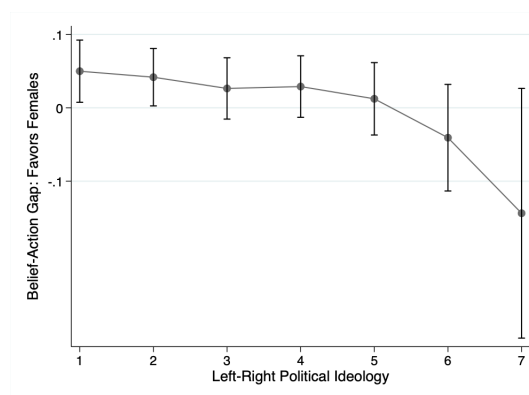


Notes: Each dot represents the regression estimated difference between hiring managers' and clients' likelihood of hiring a female worker in the *Control* condition. The coefficient is specifically on the variable $HiringManager_i$ from an OLS regression of the decision to hire a female worker in a given pair (Y_i , the dependent variable) on an indicator that the decision-maker was a hiring manager ($HiringManager_i = 1$) rather than a client ($HiringManager_i = 0$), controlling for workers' relative scores on Quiz A and highest level of education, with standard errors clustered by participant. I run this regression on pooled observations from clients and hiring managers, but only those in the control groups. This is estimated separately for different worker pairs, such that: (i) one worker strictly dominates (based on highest education and Quiz A score), (ii) one worker weakly dominates, (iii) one worker is higher on one signal and lower on the other, and (iv) where worker signals are equal. Lines represent 95% confidence intervals of the estimated coefficients.

Figure E24: Controlled Experiment: The Belief-Action Gap by Gender and Political Ideology



(a) Difference by State Election Result



(b) Difference by “Belief Shock”

Notes: Panel a) shows the mean values and 95% confidence intervals for the difference between hiring managers’ incentivized beliefs of the proportion of female workers clients selected, and the proportion of female workers that they themselves selected. This is shown separately for male and female hiring managers. Positive values indicate hiring more female workers than they believe clients select, and negative values indicate hiring less females than their beliefs suggest. Panel b) shows the same means and confidence intervals for hiring managers at each level of left-right political ideology.

F Controlled Experiment - Additional Tables

Table F13: Controlled Experiment: Hiring Manager Sample Descriptive Statistics

Variable	(1) Control		(2) Clients as-is		(3) Clients Debiased		(1)-(2)	T-test P-value	
	N	Mean/SD	N	Mean/SD	N	Mean/SD		(1)-(3)	(2)-(3)
Age	91	41.16 (15.02)	114	39.70 (14.26)	95	40.20 (12.82)	0.48	0.64	0.79
Male	91	0.59 (0.49)	114	0.48 (0.50)	95	0.42 (0.50)	0.11	0.02**	0.38
White	91	0.70 (0.46)	114	0.64 (0.48)	95	0.64 (0.48)	0.34	0.38	0.98
Black or African-American	91	0.12 (0.33)	114	0.20 (0.40)	95	0.19 (0.39)	0.12	0.20	0.82
Other Race	91	0.19 (0.39)	114	0.18 (0.38)	95	0.19 (0.39)	0.83	0.96	0.79
College Graduate	91	0.63 (0.49)	114	0.59 (0.49)	95	0.65 (0.48)	0.58	0.71	0.34
Has Advanced Degree	91	0.15 (0.36)	114	0.18 (0.38)	95	0.21 (0.41)	0.68	0.32	0.52
Left-Right Political Scale	91	3.10 (1.46)	114	3.11 (1.75)	94	3.04 (1.70)	0.95	0.81	0.77
Has Real Hiring Experience	91	0.38 (0.49)	114	0.39 (0.49)	95	0.43 (0.50)	0.88	0.52	0.59
F-test of joint significance (F-stat)							0.63	0.22	0.98
F-test, number of observations							205	185	208

Notes: This table gives mean values (and standard deviations in brackets) for each variable for hiring managers separately in each treatment condition. The values displayed for t-tests and the F-test are p-values. ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

Table F14: Controlled Experiment: Client Sample Descriptive Statistics

Variable	N	(1) Control Mean/SD	(2) Gender Score Info N Mean/SD	T-test P-value (1)-(2)
Age	102	38.25 (12.94)	100 38.95 (12.49)	0.70
Male	102	0.46 (0.50)	100 0.49 (0.50)	0.68
White	102	0.74 (0.44)	100 0.76 (0.43)	0.69
Black or African-American	102	0.13 (0.34)	100 0.15 (0.36)	0.64
Other Race	102	0.15 (0.36)	100 0.10 (0.30)	0.31
College Graduate	102	0.64 (0.48)	100 0.53 (0.50)	0.12
Has Advanced Degree	102	0.21 (0.41)	100 0.08 (0.27)	0.01**
Left-Right Political Scale	101	3.25 (1.72)	97 3.48 (1.63)	0.32
Has Real Hiring Experience	102	0.47 (0.50)	100 0.47 (0.50)	0.99
F-test of joint significance (F-stat)				0.32
F-test, number of observations				198

Notes: This table gives mean values (and standard errors in brackets) for each variable for clients separately for each treatment condition. The value displayed for t-tests and the F-test are p-values. ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

Table F15: Controlled Experiment: Regression Results: The Effect of *Clients Debaised* on Hiring Manager Beliefs about Client Choices, and on Hiring Choices

	Beliefs: Client Female Selection		Hiring Choices: Female Selection		
	(1)	(2)	(3)	(4)	(5)
Gender Score Information	0.053*** (0.019)	0.053*** (0.019)	0.054** (0.022)	0.063*** (0.021)	0.061*** (0.021)
Clients Debaised	0.085*** (0.021)	0.081*** (0.021)	0.084*** (0.025)	0.080*** (0.024)	0.075*** (0.023)
Quiz A Score Difference				-0.183*** (0.004)	-0.183*** (0.004)
Education Difference				-0.134*** (0.009)	-0.135*** (0.009)
Conditioned on Signal	-	-		X	X
All Controls Included		X			X
Observations	300	299	3190	3190	3180

Notes: Columns 1 and 2 include the results of OLS regressions of hiring managers' incentivized beliefs about the proportion of female workers that clients selected (the dependent variable). Columns 3-5 include results of regressions of a binary variable of whether a female worker was hired (the dependent variable) in a given decision for a mixed-gender worker pair, with standard errors clustered at the hiring manager level. Across models, the dependent variables are regressed on a binary indicator for whether hiring managers saw workers' Quiz B score distribution, broken down by worker gender ("Gender Score Information"; this includes all managers assigned to either the *Clients as-is* or the *Clients Debaised* conditions), and on a binary indicator for whether managers were assigned to the *Clients Debaised* condition (the primary manipulation of beliefs about client discrimination). In columns 4 and 5, regressions include controls for workers' relative productivity signals in each pair (i.e., their relative highest education level and score on Quiz A; coefficients reported), and in columns 2 and 5, regressions include additional controls for manager race, gender, age, age squared, highest education level, and political ideology. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table F16: Controlled Experiment: Instrumental Variable Regression Results: Causal Effect of Third-Party Beliefs on Hiring Choices

	(1)	(2)	(3)	(4)	(5)	(6)
Beliefs: Client Female Selection (Incentivized)	1.045*** (0.361)	0.993*** (0.343)	0.968*** (0.340)			
Beliefs: Client Female Selection (Likert)				0.520*** (0.151)	0.494*** (0.146)	0.448*** (0.136)
Gender Score Information	-0.005 (0.041)	0.006 (0.039)	0.008 (0.037)	0.007 (0.032)	0.018 (0.031)	0.022 (0.029)
Quiz A Score Difference		-0.183*** (0.004)	-0.183*** (0.004)		-0.182*** (0.004)	-0.182*** (0.004)
Education Difference		-0.135*** (0.009)	-0.135*** (0.009)		-0.133*** (0.009)	-0.134*** (0.009)
Conditioned on Signal		X	X		X	X
All Controls Included			X			X
Belief type	Incentivized	Incentivized	Incentivized	Likert	Likert	Likert
Observations	3190	3190	3180	3190	3190	3180

Notes: All columns present regression results for two-stage-least squares (2SLS) regressions of whether a female worker was hired (the dependent variable) in a given decision for a mixed-gender worker pair, with standard errors clustered at the hiring manager level. Across all columns, we test the causal effect of hiring manager beliefs about client discrimination (i.e., female worker selection) on hiring managers' own discrimination, using random assignment to the *Clients Debaised* condition as an instrument for beliefs about client discrimination. In columns 1-3, I use hiring managers' incentivized beliefs about the proportion of workers' clients selected as the main independent variable (i.e., measure of beliefs about client discrimination). In columns 4-6 I use their response to a 5-point Likert question as a measure of beliefs about client discrimination. Across models, I control for whether hiring managers saw workers' Quiz B score distribution, broken down by worker gender ("Gender Score Information"; this includes all managers assigned to either the *Clients as-is* or the *Clients Debaised* conditions). In columns 2-3 and 5-6, regressions include controls for workers' relative productivity signals in each pair (i.e., their relative highest education level and score on Quiz A; coefficients reported), and in columns 3 and 6, regressions include additional controls for manager race, gender, age, age squared, highest education level, and political ideology. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table F17: Controlled Experiment: Regression Results for Clients' Beliefs about Female-Male Worker Productivity and Choices

	Beliefs: Female - Male Scores		Choices: Female Offers		
	(1)	(2)	(3)	(4)	(5)
Gender Score Information	0.414** (0.163)	0.421*** (0.158)	0.093*** (0.021)	0.094*** (0.020)	0.100*** (0.020)
Quiz A Score Difference				-0.179*** (0.005)	-0.179*** (0.005)
Education Difference				-0.118*** (0.009)	-0.119*** (0.010)
Conditioned on Signal	-	-		X	X
All Controls Included		X			X
Observations	202	198	2158	2158	2114

Notes: Columns 1 and 2 include the results of OLS regressions of the mean difference between clients' predicted Quiz B score for female workers and their predicted Quiz B score for male workers (the dependent variable). Columns 3-5 include results of regressions of a binary variable of whether a female worker was selected in a given decision or worker pair (the dependent variable), with standard errors clustered at the client level. Across models, the dependent variables are regressed on a binary indicator for whether clients saw workers' Quiz B score distribution, broken down by worker gender (i.e., were assigned to the "Gender Score Information" condition). In columns 4 and 5, regressions include controls for workers' relative productivity signals in each pair (i.e., their relative highest education level and score on Quiz A; coefficients reported), and in columns 2 and 5, regressions include additional controls for manager race, gender, age, age squared, highest education level, and political ideology. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.